

Security of Fixed-Weight Repetitions of Special-Sound Multi-Round Interactive Proofs

Michele Battagliola¹, Riccardo Longo², Federico Pintore³, Edoardo Signorini^{4,5},
and Giovanni Tognolini³

¹ Università Politecnica delle Marche, Ancona, Italy

² Fondazione Bruno Kessler, Center for Cybersecurity, Trento, Italy

³ Università di Trento, Trento, Italy

⁴ Telsy, Turin, Italy

⁵ Politecnico di Torino, Turin, Italy

Abstract. Interactive proofs are a cornerstone of modern cryptography and, as such, used in many areas, from digital signatures to multi-party computation. Often the knowledge error κ of an interactive proof is not small enough and thus needs to be reduced. This is usually achieved by repeating the interactive proof in parallel t times. Recently, it was shown that the t -fold parallel repetition of any $(k_1, \dots, k_\mu)$ -special-sound multi-round public-coin interactive proof reduces the knowledge error from κ to κ^t , which is optimal. However, parallel repetitions lead to an increase in transcript size. A common technique to mitigate this drawback, which is often employed in digital signatures obtained by using the Fiat-Shamir transform, is to use fixed-weight challenges, i.e. vectors of challenges having a constant number of entries (for which the last component is) equal to a fixed value. While widely used, this method has not been fully assessed from a security standpoint. In particular, the effect of the technique on the knowledge error of repeated interactive proofs has remained unstudied. In this work, we fill the gap and prove that a fixed-weight repetition of a $(k_1, \dots, k_\mu)$ -special-sound multi-round public-coin interactive proof is still knowledge sound. We provide an explicit bound for the knowledge error of the protocol, proving that it matches the maximum cheating probability of a dishonest prover. Our results apply to some recently-proposed digital signatures which are supposed to be quantum resistant, for example the code-based signature CROSS.

1 Introduction

Interactive Proofs. An interactive proof for a binary relation $R \subseteq \{0, 1\}^* \times \{0, 1\}^*$ allows a prover $\mathcal{P}$ to convince a verifier $\mathcal{V}$ that a statement x admits a witness w , i.e. $(x, w) \in R$, or even that they know a witness. It is standard to require an interactive proof to be complete and sound⁶.

⁶ A verifier $\mathcal{V}$ accepts the proof produced by a honest prover $\mathcal{P}$ – having in input $(x, w) \in R$ – with *high* probability, and a verifier $\mathcal{V}$ rejects proofs for a statement x which does not admit any witness with *high* probability, respectively.

When an interactive proof is meant to allow a prover to convince a verifier they know a witness, it is further required to be knowledge sound. Informally, this means that any dishonest prover who does not know a witness can only convince a verifier with some *small* probability κ , which is called the knowledge error. This property is formalised by requiring that there exists an efficient algorithm - the extractor - that, given oracle access to a dishonest prover who succeeds with probability $\epsilon > \kappa$, outputs a witness with probability at least $\epsilon - \kappa$ up to a multiplicative polynomial loss in the security parameter.

In practice, it is usually easier to verify that the interactive proof is special sound, since this implies knowledge soundness. A 3-round interactive proof is 2-special-sound if there exists an efficient algorithm that, given two valid transcripts (a, c, z) and (a, c', z') relative to the same statement x - with the same first message a from $\mathcal{P}$ (commitment), distinct second messages (challenges) $c \neq c'$ and corresponding responses z, z' - outputs a witness w for x . This property can be generalised to a different number of input transcripts and to $(2\mu + 1)$ -round interactive proofs, leading to the notion of $(k_1, \dots, k_\mu)$ -special soundness, which coincides with the standard 2-special-soundness notion when $\mu = 1$ and $k_1 = 2$.

Often the knowledge error κ of an interactive proof is not small enough security-wise, and thus needs to be reduced. This is usually achieved by repeating the interactive proof in parallel t times. Recently, Attema and Fehr [AF22] proved that t parallel repetitions of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proof reduces the knowledge error from κ to κ^t , which is optimal.

Digital signatures. With the threat of quantum computers looming ever closer, the cryptographic community has reacted by developing alternative cryptographic solutions supposed to be resistant even to quantum algorithms. This collective effort has been further invigorated by the NIST call for standardisation [NIS17]. While the first standards covering key encapsulation and signatures have already been made public, the situation with the latter is not considered fully satisfactory, so NIST has launched an “on-ramp” process to standardise new signature schemes [NIS23]. Looking at the numerous signatures submitted to the NIST calls, two main generic design techniques stand out: the hash-and-sign and the Fiat-Shamir constructions.

Introduced by Fiat and Shamir in [FS87], the Fiat-Shamir transform allows to turn any public-coin⁷ interactive proof into a non-interactive proof and, consequently, into a digital signature. Informally, the Fiat-Shamir transform replaces random challenges sent by the verifier with outputs of a hash function, thus removing the need for interaction. Many post-quantum digital signatures have been designed exploiting this paradigm [Bar+21; Bal+23; Cho+23; BKV19; De+20; Duc+18].

It has proven to be the only viable option for isogeny-based cryptography and, arguably, the most conservative one security-wise for code-based cryptography, as testified by the second-round candidates of the NIST competition.

⁷ All verifier’s random choices are made public.

Fiat-Shamir digital signatures *inherit* the main security properties of the starting interactive proof, which almost always enjoys special soundness. In particular, the security of the resulting signature relates to the knowledge error determined by the special soundness. However, many starting interactive proofs, with notable exceptions like SQISign [De +20], only have very small (often binary) challenge spaces, which result in a big knowledge error. This is therefore one of those cases where the knowledge error must be decreased, for example by performing t parallel repetitions of the base interactive proof.

The fixed-weight optimisation. Parallel repetition of the base interactive proof significantly impacts the efficiency of the resulting scheme. A few generic techniques have been proposed to mitigate this issue. Some of them aim at limiting the number t of repetitions (e.g. the multiple-public-key optimisation, like in [BKV19; DG19]), obtaining an improvement in both execution time and transcript size at the cost, for example, of bigger public keys. Other techniques, on the other hand, aim at reducing the transcript size at the cost of a slight increase in execution time. This is particularly desired when storage or transmission latency are the main concern. Among the latter techniques, one of the most common optimisations is the *fixed-weight optimisation*⁸, which uses *fixed-weight* challenge vectors.

The challenges are the random coins sent by the verifier to the prover. In particular, for a single execution of a $(2\mu + 1)$ -round interactive proof, they are the elements of a finite challenge space $\text{Ch} = \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$. The t -fold parallel repetition expands the challenge space from Ch to Ch^t . Within the fixed-weight optimisation, are considered only challenges $\mathbf{c} = ((\mathbf{c})_1, \dots, (\mathbf{c})_t) \in \text{Ch}^t$, where $(\mathbf{c})_j \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$, such that $((\mathbf{c})_1^{[1]}, \dots, (\mathbf{c})_t^{[1]})$ has a constant number w of entries equal to a fixed value. The value w is called weight.

This optimisation has proven to be particularly helpful when different challenges have drastically-different response sizes [Bal+23; GPV24; Bar+21; Cho+23; Tan+22; Beu+23; BKP20]. A typical example of this scenario is when responses for a challenge are structured mathematical objects (e.g. matrices or points on elliptic curves) while those for the other(s) are seeds used for randomness, that are significantly shorter.

For example, the isogeny-based digital signature in [GPV24] is obtained by applying the Fiat-Shamir transform to a 3-round 3-special-sound interactive proof with challenge space $\text{Ch} = \{-1, 0, 1\}$. After applying three other optimisations, the size of the responses for the three challenges are of $r_1 = 384$, $r_2 = 475$ and $r_3 = 1737$ bits, respectively, for the set of parameters targeting 128 bits of security. The fixed-weight optimisation then reduces the (average) size of a signature from $s_1 = 188771$ bits [GPV24, Table 4] down to $s_2 = 170263$ bits, with a 9.81% gain⁹.

⁸ This optimisation is sometimes referred as the one with *unbalanced* challenge vectors.

⁹ In particular $s_1 = 128 + 218 \cdot (r_1 + r_2 + r_3)/3$ and $s_2 = 128 + 48 \cdot r_3 + 202 \cdot (r_1 + r_2)/2$, using 250 and 48 for t and the weight w , respectively, as per [GPV24]. These figures differ from those in [GPV24, Table 5], where gains are considered with respect to the base interactive proof.

Here we note that, for each specific scheme and targeted security level, the pair (t, w) which guarantees the maximum gain is determined by an exhaustive search on all *small* values of w and the corresponding t which lead to an appropriate dimension of the overall challenge space.

The security of the fixed-weight optimisation is well understood in the case of a 3-round, public-coin, 2-special-sound interactive proof, i.e. $\mu = 1$ and $k_1 = 2$. In fact, in this case the 2-special soundness of the base interactive proof is preserved by its fixed-weight repetitions. However, the picture becomes fuzzy when $\mu = 1$ and $k_1 > 2$, and even more when $\mu > 1$. In other words, already in the case of k -special-sound 3-round public-coin interactive proofs with $k > 2$ it is not clear whether the fixed-weight t -fold parallel repetition satisfies any useful notion of special soundness. In light of this and the implications it would have on the provable security of existing signatures like CROSS [Bal+23] (that was recently included by NIST in the fourteen Second Round Candidates for standardization) or [GPV24] and future protocols, the following research question naturally arises:

Does a fixed-weight repetition of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round public-coin interactive proof enjoy knowledge soundness?

Our Contribution. Building on the results from [AF22] we positively answer the question above by explicitly building a knowledge extractor and precisely bounding the knowledge error.

More in details, we prove that the t -fold repetition with fixed weight w of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proof is knowledge sound, and we provide an explicit expression for the knowledge error. Such knowledge error coincides with the maximum cheating probability of a dishonest prover, which is directly derived from the maximum size of the set of challenges to which the prover can answer without actually knowing a witness. This shows that our result is optimal and, in addition, it allows to formally prove the security of the interactive proofs underlying some recent post-quantum signatures, such as the code-based signature CROSS [Bal+23] and the SIDH-based signature of [GPV24].

Technical Overview. One of the main results of [AF22] is a knowledge extractor $\mathcal{E}$ for k -special-sound 3-round interactive proofs, whose success probability when applied to a dishonest prover $\mathcal{P}^*$ can be expressed in terms of a novel characterization of the power of $\mathcal{P}^*$. Specifically, the ability of $\mathcal{P}^*$ to correctly answer to a random challenge is measured by $\delta_k(\mathcal{P}^*)$, its worst-case success probability when $k - 1$ challenges are removed from the challenge space. This new framework is particularly convenient when moving to parallel repetitions of the interactive proof. In fact, starting from a dishonest prover $\mathcal{P}^*$ against the t -parallel repetition of a k -special-sound 3-round interactive proof, it is possible to build t provers $\mathcal{P}_1^*, \dots, \mathcal{P}_t^*$ against the single instance of the interactive proof. By applying the previous extractor in parallel to the provers of the single instance, the probability of witness extraction can be expressed via $\delta_k(\mathcal{P}_1^*) + \dots + \delta_k(\mathcal{P}_t^*)$. From this, an optimal bound on the knowledge error of the parallel repetition is obtained.

The extraction algorithm $\mathcal{E}$ of [AF22] queries the dishonest prover on uniformly-sampled challenges. Instead, when we consider fixed-weight repetitions, challenges must be sampled according to a different distribution. At the core of our result is therefore a generalisation of the knowledge extractor from [AF22] which allows for the sampling of challenges according to an arbitrary distribution $\mathcal{D}$ over the challenge space. More in detail, we show that the extraction probability is given by $\delta_k(\mathcal{P}^*, \mathcal{D})/k$ for a k -special-sound 3-round interactive proof, where the probability space is defined by the challenges being sampled according to $\mathcal{D}$. For fixed-weight repetitions, we can then apply a similar approach as before: starting from a dishonest prover for the fixed-weight t -fold repetition of the interactive proof, we build t dishonest provers on the single instance of the interactive proof. By applying the generalised extractor in parallel, we obtain a bound on the knowledge error of the parallel repetition of the interactive proof (Section 3). A similar strategy is then applied to prove knowledge soundness of fixed-weight repetitions of generic $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proofs. In particular, we first generalise the multi-round extractor from [AF22] over arbitrary distributions and then apply it to fixed-weight repetitions (Section 5).

Although the obtained bounds cannot be expressed directly in terms of the knowledge error of a single instance, the obtained knowledge errors coincide with the maximum cheating probability of a dishonest prover, meaning that our result is optimal. As already observed, the maximum cheating probability of a dishonest prover is derived from the maximum size of the sets of challenges to which the prover can answer without actually knowing a witness. We have translated the problem of computing these sizes into finding upper bounds for the cardinality of particular subsets of the Cartesian product of finite sets which satisfy some conditions on the components. In particular, we compute the maximum size of a set of sequences when we limit the number of different values that may appear in any single component (Proposition 1 and Section 4). These combinatorial results, from which a bound on the knowledge error of fixed-weight repetitions is deduced, may be of independent interest and find application in independent scenarios, and so we have formulated and proved them in full generality.

Organisation. In Section 2 we provide some preliminaries and definitions on interactive proofs, with a focus on multi-round ones. Section 3 tackles the security of the fixed-weight optimization in the easier case of 3-round interactive proofs. This section serves also to introduce in a simpler setting the techniques that will be used in the multi-round case. Next, in Section 4 we discuss some combinatorial results that, while potentially interesting on their own, will be essential for Section 5. Section 5 contains the core of our work and deals with the general multi-round case. Lastly, in Section 6 we identify some applications and future research directions.

2 Preliminaries

Notation. We denote by $\mathbb{N}^*$ the set of non-zero natural numbers. For a finite set X , we write $|X|$ for its cardinality and by $|x|$ we denote the number of bits

necessary to represent an element $x \in X$. We denote by $\{0, 1\}^*$ the set of binary strings of arbitrary length.

When s is a list or a vector, we write $(s)_i$ to denote the i -th element of s . If S is a set whose elements are lists or vectors, we define $(S)_i := \{x : \exists s \in S : (s)_i = x\}$.

Given μ finite sets $\text{Ch}^{[1]}, \dots, \text{Ch}^{[\mu]}$ and $\mathbf{c} \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$, we will write $\mathbf{c} = (c^{[1]}, \dots, c^{[\mu]})$ where $c^{[i]} \in \text{Ch}^{[i]}$ for all $i \in \{1, \dots, \mu\}$. Furthermore, given $t \in \mathbb{N}^*$ and $\mathbf{c} \in (\text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]})^t$, we will write $\mathbf{c} = ((\mathbf{c})_1, \dots, (\mathbf{c})_t)$ where $(\mathbf{c})_j \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$ and $(\mathbf{c})_j^{[i]} \in \text{Ch}^{[i]}$ for all $j \in \{1, \dots, t\}, i \in \{1, \dots, \mu\}$.

Interactive Proofs. The aim of this work is proving the knowledge soundness of fixed-weight repetitions of $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round public-coin interactive proofs. In this section we recall the definition of interactive proof, some related notions and usual security requirements. Moreover, we describe the fixed-weight repetition of an interactive proof.

Definition 1 (Binary relation). A binary relation is a finite set $R \subseteq X \times Y$, where $X, Y \subseteq \{0, 1\}^*$. Given $(x, y) \in R$, we say that y is a witness for the statement x . The set $L_R = \{x \in X \mid \exists y \in Y \text{ s.t. } (x, y) \in R\}$ is called the set of true statements for R , or its language.

Definition 2 (Interactive Proof). An interactive proof $(\mathcal{P}, \mathcal{V})$ for a binary relation $R \subseteq X \times Y$ is an interactive protocol between two probabilistic polynomial-time machines $\mathcal{P}$ and $\mathcal{V}$. The prover $\mathcal{P}$ takes as input a pair $(x, y) \in R$ while the verifier $\mathcal{V}$ takes as input x . As the output of the protocol - denoted by $(\mathcal{P}(y), \mathcal{V})(x)$ - $\mathcal{V}$ either accepts (outputs 1) or rejects (outputs 0). We say that a transcript, i.e. the set of all messages exchanged in a protocol execution, is accepting (rejecting) if $\mathcal{V}$ accepts (rejects, respectively).

Throughout this work we assume that, within an execution of an interactive proof $(\mathcal{P}, \mathcal{V})$, the prover $\mathcal{P}$ always sends the first and the last message. Hence, the number of communication rounds is odd, i.e. of the form $2\mu + 1$ with $\mu \in \mathbb{N}^*$. We refer to an interactive proof having $2\mu + 1$ communication rounds with the name $(2\mu + 1)$ -round interactive proof. When $\mu = 1$, and thus the rounds are only 3, we call it *Sigma protocol*.

Definition 3 (Public-Coin). An interactive proof $(\mathcal{P}, \mathcal{V})$ is public-coin if all $\mathcal{V}$'s random choices are made public.

If an interactive proof is public-coin, the verifier needs to send to the prover only their random choices. For this reason, we call *challenges* the messages sent by the verifier and *challenge set* the set from which verifier's messages are sampled. In the case of a $(2\mu + 1)$ -round interactive proof, we define the challenge set Ch as the Cartesian product of μ round challenge sets $\text{Ch}^{[i]}$, with $i \in \{1, \dots, \mu\}$, meaning that the challenge for the i -th round is sampled from $\text{Ch}^{[i]}$.

Commonly, an interactive proof is required to satisfy completeness and soundness, as per definitions below.

Definition 4 (Completeness). An interactive proof $(\mathcal{P}, \mathcal{V})$ for a binary relation $R \subseteq X \times Y$ is complete if, for every $(x, y) \in R$, we have:

$$\Pr[(\mathcal{P}(y), \mathcal{V})(x) = 0] \leq \rho(x)$$

where the value $\rho(x)$ - called completeness error - is negligible (in $|x|$). If $\rho(x) = 0$ for all $x \in L_R$, the protocol is said to be perfectly complete.

Definition 5 (Soundness). An interactive proof $(\mathcal{P}, \mathcal{V})$ for a binary relation $R \subseteq X \times Y$ is sound if, for every $x \notin L_R$ and a (potentially-dishonest) prover $\mathcal{P}^*$, we have:

$$\Pr[(\mathcal{P}^*, \mathcal{V})(x) = 1] \leq \sigma(x)$$

where the value $\sigma(x)$ - called soundness error - is negligible (in $|x|$).

We note that an interactive proof which satisfies both the previous properties allows a prover $\mathcal{P}$ to convince a verifier $\mathcal{V}$ that a statement x is true. It does not guarantee anything about $\mathcal{P}$'s knowledge of a witness y such that $(x, y) \in R$. This stronger feature requires *knowledge soundness*.

Definition 6 (Knowledge Soundness). An interactive proof $(\mathcal{P}, \mathcal{V})$ for a binary relation $R \subseteq X \times Y$ is knowledge sound, with knowledge error κ , if there exists an algorithm $\mathcal{E}$ that, given as input any $x \in X$ and rewindable oracle access to a (potentially-dishonest) prover $\mathcal{P}^*$, runs in an expected polynomial time (in $|x|$) and outputs a witness $y \in Y$ for x with probability:

$$\Pr[(x, \mathcal{E}^{\mathcal{P}^*}(x)) \in R] \geq \frac{\varepsilon(x, \mathcal{P}^*) - \kappa(x)}{\text{poly}(|x|)},$$

where $\varepsilon(x, \mathcal{P}^*) = \Pr[(\mathcal{P}^*, \mathcal{V})(x) = 1]$. The algorithm $\mathcal{E}$ is called knowledge extractor.

Remark 1. As noted in [AF22, Remark 1], it is enough to prove that the Knowledge Soundness property of Definition 6 holds for any deterministic prover $\mathcal{P}^*$. More precisely, suppose that $\mathcal{P}^*$ is a probabilistic prover, and denote by $\mathcal{P}^*[r]$ the deterministic prover obtained by setting the randomness of $\mathcal{P}^*$ to r . Then, it is easy to show that $\varepsilon(x, \mathcal{P}^*) = \mathbb{E}[\varepsilon(x, \mathcal{P}^*[r])]$ and $\Pr[(x, \mathcal{E}^{\mathcal{P}^*}(x)) \in R] = \mathbb{E}[\Pr[(x, \mathcal{E}^{\mathcal{P}^*[r]}(x)) \in R]]$, where the expected value is taken over the random choice of r . For this reason, to simplify the subsequent analysis we will assume that the prover $\mathcal{P}^*$ is deterministic, unless otherwise specified.

Definition 7 (Proof of Knowledge). An interactive proof $(\mathcal{P}, \mathcal{V})$ for a binary relation $R \subseteq X \times Y$ which satisfies both completeness with completeness error ρ and knowledge soundness with knowledge error κ is a proof of knowledge if there exists a positive-definite polynomial p over the integers such that $1 - \rho(x) \geq \kappa(x) + \frac{1}{p(|x|)}$ for all $x \in X$.

A common strategy to prove the knowledge soundness of a public-coin interactive proof is showing that it enjoys special soundness, which means, informally, that there exists an extracting algorithm able to compute a witness given enough accepting transcripts relative to a true statement x . While the definition for Sigma protocols can be simply stated, for the general $(2\mu + 1)$ -round case we need to firstly introduce the notion of tree of transcripts.

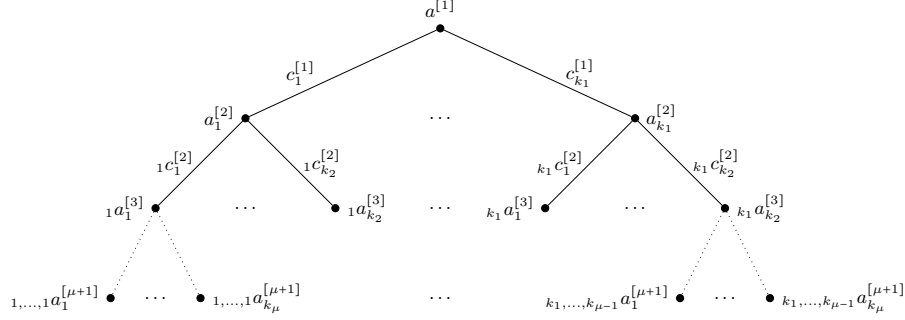


Fig. 1. Graphical representation of a $(k_1, \dots, k_\mu)$ -tree of transcripts for a $(2\mu + 1)$ -round public-coin interactive proof. Left subscripts represent the ancestor nodes, superscripts represent the corresponding round, while right subscripts are used to enumerate edges originating from a node and their corresponding arrival nodes.

Definition 8 (Tree of Transcripts). Let $k_1, \dots, k_\mu, N_1, \dots, N_\mu \in \mathbb{N}^*$, $R \subseteq X \times Y$ be a binary relation and $(\mathcal{P}, \mathcal{V})$ a $(2\mu + 1)$ -round public-coin interactive proof for R , where $\mathcal{V}$ samples i -th challenges ($i \in \{1, \dots, \mu\}$) from a set $\text{Ch}^{[i]}$ of cardinality $N_i \geq k_i$. A $(k_1, \dots, k_\mu)$ -tree of transcripts for $(\mathcal{P}, \mathcal{V})$ is a set of $K = \prod_{i=1}^\mu k_i$ transcripts relative to a given statement $x \in X$, arranged in the following tree structure, where nodes correspond to prover's messages while edges to verifier's challenges. From every node at level i , with $i \in \{1, \dots, \mu\}$, exactly k_i edges originate, corresponding to k_i pairwise-distinct challenges belonging to $\text{Ch}^{[i]}$. Then, each of the K transcripts corresponds to exactly one path from the root node to a leaf node.

A graphical representation of a tree of transcripts is provided in Figure 1, where $a^{[1]}$ denotes the prover's first message, $c_1^{[1]}, \dots, c_{k_1}^{[1]}$ are sampled from $\text{Ch}^{[1]}$, and so on.

Definition 9 $((k_1, \dots, k_\mu)$ -Special Soundness). Let $k_1, \dots, k_\mu, N_1, \dots, N_\mu \in \mathbb{N}^*$ and $R \subseteq X \times Y$ be a binary relation. A $(2\mu + 1)$ -round public-coin interactive proof $(\mathcal{P}, \mathcal{V})$ for R , where $\mathcal{V}$ samples i -th challenges ($i \in \{1, \dots, \mu\}$) from a set $\text{Ch}^{[i]}$ of cardinality $N_i \geq k_i$, is $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ special sound, or simply $(k_1, \dots, k_\mu)$ -special sound, if there exists a polynomial-time algorithm that, on input a true statement $x \in X$ and a $(k_1, \dots, k_\mu)$ -tree of accepting transcripts for $(\mathcal{P}, \mathcal{V})$ and relative to x , outputs a witness $y \in Y$ for x .

In the case of a Sigma protocol, it is well-known that k -out-of- N -special soundness implies knowledge soundness with knowledge error $(k-1)/N$ [HL10]. The general $(2\mu+1)$ -round case is much more involved, and it has only recently been shown [ACK21] that $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ -special soundness tightly implies knowledge soundness, with knowledge error:

$$\kappa = 1 - \prod_{i=1}^{\mu} \frac{(N_i - k_i + 1)}{N_i}.$$

A common solution to decrease the knowledge error of a $(2\mu+1)$ -round knowledge-sound interactive proof $(\mathcal{P}, \mathcal{V})$ is to repeat it in parallel multiple times, i.e. the prover and the verifier run t parallel executions of the protocol and the verifier accepts if the resulting t transcripts are accepting. We denote by $(\mathcal{P}^t, \mathcal{V}^t)$ the t -fold parallel repetition of $(\mathcal{P}, \mathcal{V})$. While this technique has been broadly adopted, it was only in 2022 that Attema and Fehr [AF22] proved that the t -fold parallel repetition of any $(k_1, \dots, k_\mu)$ -special-sound $(2\mu+1)$ -round public-coin interactive proof optimally reduces the knowledge error from κ down to κ^t .

Fixed-Weight Repetition. Parallel repetition of knowledge-sound interactive proofs improves security at the price of bigger transcripts. When responses to different challenges have very unbalanced sizes and compactness is a bigger concern than computational efficiency, it can be beneficial to use *fixed-weight challenges*.

Definition 10 (Weight). Let Ch be a finite set, $t \in \mathbb{N}^*$ and $\tilde{c} \in \text{Ch}$. For an element $c = ((c)_1, \dots, (c)_t) \in \text{Ch}^t$, we define the weight of c with respect to $\tilde{c}$ as:

$$\text{wt}_{\tilde{c}}(c) := |\{j \in \{1, \dots, t\} : (c)_j = \tilde{c}\}|.$$

Definition 11. Let $t, w, \mu \in \mathbb{N}^*$ be such that $t \geq w$, let $\text{Ch}^{[1]}, \dots, \text{Ch}^{[\mu]}$ be finite sets and let $\tilde{c} \in \text{Ch}^{[1]}$. Given $\text{Ch} = \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$, we denote by $\text{Ch}_{\tilde{c}}^{t,w}$ the set of elements $\mathbf{c} \in \text{Ch}^t$ for which $\text{wt}_{\tilde{c}}((\mathbf{c})_1^{[\mu]}, \dots, (\mathbf{c})_t^{[\mu]}) = w$, i.e.:

$$\text{Ch}_{\tilde{c}}^{t,w} := \left\{ \mathbf{c} \in \text{Ch}^t : \text{wt}_{\tilde{c}}\left(\left((\mathbf{c})_1^{[\mu]}, \dots, (\mathbf{c})_t^{[\mu]}\right)\right) = w \right\}.$$

When $\tilde{c}$ is clear from the context, we will simplify the notation and write $\text{Ch}^{t,w}$ instead of $\text{Ch}_{\tilde{c}}^{t,w}$. Furthermore, when Ch is not a Cartesian product but a simple set (i.e., $\mu = 1$ and so $\text{Ch} = \text{Ch}^{[1]}$), we will simply denote by $\text{Ch}_{\tilde{c}}^{t,w}$ the set:

$$\{c = ((c)_1, \dots, (c)_t) \in \text{Ch}^t : \text{wt}_{\tilde{c}}(c) = w\}.$$

Definition 12 (Fixed-weight Repetition). Let $k_1, \dots, k_\mu, N_1, \dots, N_\mu \in \mathbb{N}^*$, $R \subseteq X \times Y$ be a binary relation and $(\mathcal{P}, \mathcal{V})$ be a $(2\mu+1)$ -round public-coin interactive proof for R , where $\mathcal{V}$ samples i -th challenges ($i \in \{1, \dots, \mu\}$) from a set $\text{Ch}^{[i]}$ of cardinality $N_i \geq k_i$. Therefore, the challenge set of $(\mathcal{P}, \mathcal{V})$ is $\text{Ch} = \prod_{i=1}^{\mu} \text{Ch}^{[i]}$. Let $\tilde{c}$ be a given element of $\text{Ch}^{[1]}$. A (t, w) -fixed-weight parallel repetition of $(\mathcal{P}, \mathcal{V})$ with respect to $\tilde{c}$, which we denote by $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$, is a t -fold parallel repetition of $(\mathcal{P}, \mathcal{V})$ whose challenge set is $\text{Ch}_{\tilde{c}}^{t,w}$.

Throughout this work, we will consider fixed-weight repetitions only for $(2\mu + 1)$ -round public-coin interactive proofs for which there exists a unique element $\tilde{c} \in \text{Ch}^{[\mu]}$ such that, for every possible $\mathbf{c} = (c^{[1]}, \dots, c^{[\mu]}) \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$, the response size when $c^{[\mu]} = \tilde{c}$ is significantly higher than when $c^{[\mu]} \neq \tilde{c}$. Under this assumption, a fixed-weight repetition can lead to a more compact protocol compared to a plain parallel repetition, as it was shown in [GPV24; Bar+21; Bal+23; Cho+23; Tan+22].

Remark 2. In Definition 12, we choose to consider the fixed element $\tilde{c}$ as an element of $\text{Ch}^{[\mu]}$ rather than of the challenge set of previous rounds or a Cartesian product of (a subset of) them. This is consistent with the concrete instances of the fixed-weight technique that have appeared so far (see the list above).

Although the fixed-weight-repetition technique can be considered well established among cryptosystem designers, to the best of our knowledge, its knowledge soundness has not been formally investigated so far. More precisely, when $\mu = 1$ and $k_1 = 2$, 2-special soundness is trivially preserved by the optimisation technique. However, the picture becomes much fuzzier as soon as $k_1 > 2$ and, even more, when $\mu > 1$. For these cases, no formal proof of the knowledge soundness has been provided so far. In Sections 3 and 5 we fill this gap with a positive result. For the multi-round case, tackled in Section 5, some preliminary mathematical results are necessary, which will be the focus of Section 4.

3 Fixed-Weight Repetition of a k -Special-Sound Sigma Protocol

In this section we prove the security of the fixed-weight optimisation for 3-round interactive proofs, introducing also the techniques that will be used for the multi-round case in Section 5.

Let $(\mathcal{P}, \mathcal{V})$ be a k -special-sound Sigma protocol, with challenge space Ch of cardinality N . We make a two-fold assumption on the protocol:

- the knowledge error $(k - 1)/N$ is not negligible in the security parameter;
- the response size for a specific challenge $\tilde{c} \in \text{Ch}$ significantly exceeds the response sizes for the other challenges.

To reduce the knowledge error while limiting the increase in the overall response size, a common technique is to repeat the interactive proof in parallel t times, with exactly w repetitions using the unfavourable challenge $\tilde{c}$. We denote by $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ the resulting protocol. In this section, we want to prove that such scheme is knowledge sound. To this end, we first slightly generalize the notation and results in [AF22] by, at times, replacing the uniform distribution with an arbitrary one.

A dishonest prover against the Sigma protocol $(\mathcal{P}, \mathcal{V})$ can be described as an algorithm $\mathcal{A}: \text{Ch} \rightarrow \{0, 1\}^*$. Let $\mathcal{V}: \text{Ch} \times \{0, 1\}^* \rightarrow \{0, 1\}$ be a verification function.

Throughout this section $\mathcal{D}$ will denote a probability distribution over Ch with support Ch . We define the $\mathcal{D}$ -success probability of $\mathcal{A}$ as:

$$\varepsilon^V(\mathcal{A}, \mathcal{D}) = \Pr[V(C, \mathcal{A}(C)) = 1],$$

where the probability space is defined by C being sampled from Ch according to the probability distribution $\mathcal{D}$ and by the randomness of $\mathcal{A}$. When $\mathcal{D}$ is the uniform distribution over Ch , we simply write $\varepsilon^V(\mathcal{A})$. Similarly, we adapt the worst-case success probability of $\mathcal{A}$ for a random challenge when $k-1$ challenges are removed from Ch , as follows:

$$\delta_k^V(\mathcal{A}, \mathcal{D}) = \min_{S \subseteq \text{Ch}; |S|=k-1} \Pr[V(C, \mathcal{A}(C)) = 1 \mid C \notin S].$$

It is easily seen that $\delta_1^V(\mathcal{A}, \mathcal{D}) = \varepsilon^V(\mathcal{A}, \mathcal{D})$. Moreover, in the following lemma we prove that $\delta_k^V(\mathcal{A}, \mathcal{D})$ is a decreasing function in k for any choice of $\mathcal{D}$.

Remark 3. In the next two lemmas we prove some properties of δ_k^V . We will prove them only in the case of a deterministic adversary $\mathcal{A}$, however, as noted in Remark 1 it is always possible to translate them to the probabilistic case by making explicit the randomness r used by $\mathcal{A}$ and taking the expected value over all the possible r .

Lemma 1. *Let $\mathcal{D}$ be a probability distribution over Ch . Then, for all $k \in \mathbb{N}^*$:*

$$\delta_{k+1}^V(\mathcal{A}, \mathcal{D}) \leq \delta_k^V(\mathcal{A}, \mathcal{D}).$$

Proof. Let C be a random variable distributed as $\mathcal{D}$ and let $S \subseteq \text{Ch}$ be such that it minimizes $\delta_k^V(\mathcal{A}, \mathcal{D})$. Moreover, let $\bar{S} = \text{Ch} \setminus S$ and $\bar{S}' = \{c \in \bar{S} \mid V(c, \mathcal{A}(c)) = 1\}$. Now, we need to distinguish whether $\bar{S}' = \emptyset$ or not.

If $\bar{S}' \neq \emptyset$ then, for any $c' \in \bar{S}'$, we have:

$$\begin{aligned} \delta_{k+1}^V(\mathcal{A}, \mathcal{D}) &\leq \Pr[V(C, \mathcal{A}(C)) = 1 \mid C \notin S \cup \{c'\}] = \frac{(\sum_{c \in \bar{S}'} \Pr[C = c]) - \Pr[C = c']}{(\sum_{c \in \bar{S}} \Pr[C = c]) - \Pr[C = c']} \\ &\leq \frac{\sum_{c \in \bar{S}'} \Pr[C = c]}{\sum_{c \in \bar{S}} \Pr[C = c]} = \delta_k^V(\mathcal{A}, \mathcal{D}). \end{aligned}$$

where the second inequality follows by observing that $\bar{S}' \subseteq \bar{S}$, and therefore $\sum_{c \in \bar{S}'} \Pr[C = c] \leq \sum_{c \in \bar{S}} \Pr[C = c]$.

If $\bar{S}' = \emptyset$ then it means that all $c \in \text{Ch} \setminus S$ are such that $V(c, \mathcal{A}(c)) = 0$. Given $S'' \supset S$, with $|S''| = k$, it is clear that $\Pr[V(C, \mathcal{A}(C)) = 1 \mid C \notin S''] = 0$, thus $\delta_{k+1}^V(\mathcal{A}, \mathcal{D}) = 0$. So also in this case we have $\delta_{k+1}^V(\mathcal{A}, \mathcal{D}) \leq \delta_k^V(\mathcal{A}, \mathcal{D})$. $\square$

In the following, we also consider the restriction $\mathcal{D}|_S$ of $\mathcal{D}$ to a subset $S \subseteq \text{Ch}$.

Definition 13 (Distribution Restriction). *Let $\mathcal{D}$ be a probability distribution over Ch and let $X \sim \mathcal{D}$. For any subset $S \subseteq \text{Ch}$, the restriction $\mathcal{D}|_S$ of $\mathcal{D}$ to S is defined by the following density function:*

$$\Pr[X|_S = x] = \frac{\Pr[X = x]}{\sum_{x' \in S} \Pr[X = x']}, \quad \forall x \in S.$$

Input: $k \in \mathbb{N}^*$, Ch a finite set with $|\text{Ch}| = N \geq k$ and $S \subseteq \text{Ch}$ with $|S| \geq k$.
Oracle access: algorithm $\mathcal{A}: \text{Ch} \rightarrow \{0, 1\}^*$ and verification function $V: \text{Ch} \times \{0, 1\}^* \rightarrow \{0, 1\}$.
Output: if successful, $(c_1, y_1), \dots, (c_k, y_k) \in \text{Ch} \times \{0, 1\}^*$ with $V(c_i, y_i) = 1$ for all i and $c_i \neq c_j$ for $i \neq j$, otherwise $\perp$.

- 1: Sample $c_1 \in S$ according to $\mathcal{D}|_S$ and obtain $y_1 \leftarrow \mathcal{A}(c_1)$
- 2: **if** $V(c_1, y_1) = 0$ **then** abort and output $\perp$
- 3: **if** $V(c_1, y_1) = 1$ and $k = 1$ **then** output $(c_1, y_1) \in \text{Ch} \times \{0, 1\}^*$
- 4: **else**
- 5: **repeat**
- 6: set $S' = S \setminus \{c_1\}$ and run $\mathcal{E}^{\mathcal{A}}(\mathcal{D}|_{S'})$
- 7: set $\text{coin} \leftarrow V(d, \mathcal{A}(d))$ with $d \in S$ sampled according to $\mathcal{D}|_S$
- 8: **until** $\mathcal{E}^{\mathcal{A}}(\mathcal{D}|_{S'})$ outputs $(c_2, y_2), \dots, (c_k, y_k)$ or $\text{coin} = 1$
- 9: **if** $\text{coin} = 1$ **then return** $\perp$
- 10: **else return** $(c_1, y_1), \dots, (c_k, y_k)$

Fig. 2. Extractor $\mathcal{E}^{\mathcal{A}}(\mathcal{D}|_S)$.

A simple adaptation of [AF22, Lemma 2] proves the existence of an extraction algorithm $\mathcal{E}^{\mathcal{A}}(\mathcal{D})$ - with oracle access to $\mathcal{A}$ and that samples challenges from Ch following the distribution $\mathcal{D}$ - which runs in expected polynomial time and succeeds with probability at least $\delta_k^V(\mathcal{A}, \mathcal{D})/k$. The extraction algorithm is described in Figure 2.

Lemma 2. *Let $k \in \mathbb{N}^*$, let Ch be a finite set with cardinality $N \geq k$, and let $V: \text{Ch} \times \{0, 1\}^* \rightarrow \{0, 1\}$ be an arbitrary function and $\mathcal{D}$ an arbitrary probability distribution over Ch . Then there exists an algorithm $\mathcal{E}^{\mathcal{A}}(\mathcal{D})$ so that, given oracle access to any (probabilistic) algorithm $\mathcal{A}: \text{Ch} \rightarrow \{0, 1\}^*$, $\mathcal{E}^{\mathcal{A}}(\mathcal{D})$ requires an expected number of at most $2k - 1$ queries to $\mathcal{A}$ and, with probability at least $\delta_k^V(\mathcal{A}, \mathcal{D})/k$, it outputs k pairs $(c_1, y_1), (c_2, y_2), \dots, (c_k, y_k) \in \text{Ch} \times \{0, 1\}^*$ with $V(c_i, y_i) = 1$ for all i and $c_i \neq c_j$ for all $i \neq j$.*

Proof. The proof is similar to the proof of [AF22, Lemma 2]. In the original proof, the extractor samples the challenges uniformly from a subset $S \subseteq \text{Ch}$. Here, we need to consider the natural restriction of $\mathcal{D}$ to S as per Definition 13. Since we are assuming that $\mathcal{D}$ has support equal to Ch , the restriction is well-defined. Then Lemma 1 is enough to adapt the proof of [AF22, Lemma 2] and obtain the claim. $\square$

3.1 Knowledge Soundness of Fixed-Weight Repetitions

We now consider the (t, w) -fixed-weight repetition $(\mathcal{P}^{t, w}, \mathcal{V}^{t, w})$ of a k -out-of- N special-sound Sigma protocol $(\mathcal{P}, \mathcal{V})$. With reference to Definitions 10 and 12,

we assume that the challenge space for $(\mathcal{P}, \mathcal{V})$ is $\text{Ch} = \{0, \dots, N-1\}$ and the unfavourable challenge is $\tilde{c} = 0$, and we write $\text{wt}(c)$ in place of $\text{wt}_0(c)$.

The uniform distribution on $\text{Ch}^{t,w} = \text{Ch}_0^{t,w}$ induces t probability distributions $\mathcal{D}_i$ on Ch , obtained by taking the i -th component of a challenge uniformly sampled from $\text{Ch}^{t,w}$.

Definition 14. For every $i \in \{1, \dots, t\}$, we define the probability distribution $\mathcal{D}_i$ over Ch as the probability distribution having the following density function:

$$\Pr[D_i = a] = \frac{|\{c \in \text{Ch}^{t,w} \mid (c)_i = a\}|}{|\text{Ch}^{t,w}|}, \quad \forall a \in \text{Ch}.$$

An adversary against $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ is described by an algorithm $\mathcal{A}: \text{Ch}^{t,w} \rightarrow \{0, 1\}^*$. The success probability of $\mathcal{A}$ is defined as:

$$\varepsilon^V(\mathcal{A}) = \Pr[\mathbf{V}(C, \mathcal{A}(C)) = 1],$$

for some verification algorithm $\mathbf{V}: \text{Ch}^{t,w} \times \{0, 1\}^* \rightarrow \{0, 1\}$, where C is a random variable uniformly distributed over $\text{Ch}^{t,w}$.

From $\mathcal{A}$, we can construct t algorithms $\mathcal{A}_1, \dots, \mathcal{A}_t$, considering only a single invocation of $(\mathcal{P}, \mathcal{V})$. In particular, each $\mathcal{A}_i$ takes as input a challenge $c_i \in \text{Ch}$ and runs $y \leftarrow \mathcal{A}(c = (c_i, \bar{c}))$, where $\bar{c}$ is sampled uniformly at random from $\text{Ch}^{t-1, w-1}$ if $c_i = 0$ or from $\text{Ch}^{t-1, w}$ otherwise, and $\mathcal{A}$ appropriately reorder its input so that c_i is the i -th component of c (i.e. $(c)_i = c_i$). Finally, $\mathcal{A}_i$ returns y along with $\bar{c}$.

Notice that, when the input challenge c_i for $\mathcal{A}_i$ is sampled according to the probability distribution $\mathcal{D}_i$ over Ch (see Definition 14), then the inputs passed to $\mathcal{A}$ are uniformly distributed over $\text{Ch}^{t,w}$. In this case, for each $\mathcal{A}_i$, we can run the extractor $\mathcal{E}^{\mathcal{A}_i}(\mathcal{D}_i)$ of Figure 2. From Lemma 2, the extraction succeeds with probability at least $\delta_k^V(\mathcal{A}_i, \mathcal{D}_i)/k$, where:

$$\delta_k^V(\mathcal{A}_i, \mathcal{D}_i) = \min_{S_i \subset \text{Ch} \mid |S_i| = k-1} \Pr[\mathbf{V}(D_i, \mathcal{A}(D_i)) = 1 \mid D_i \notin S_i],$$

D_i is distributed as $\mathcal{D}_i$ and $\mathbf{V}$ appropriately reorder its input¹⁰.

In the following lemma we show that, when executed in parallel, at least one of the extractors $\mathcal{E}^{\mathcal{A}_i}(\mathcal{D}_i)$ succeeds with high probability in producing k challenge-response pairs that verify $\mathbf{V}$ and such that the i -th components of the challenges are all distinct.

Lemma 3. Let $k, t \in \mathbb{N}^*$, $1 \leq w \leq t$ and Ch a finite set with cardinality $N \geq k$. Let $\mathbf{V}: \text{Ch}^{t,w} \times \{0, 1\}^* \rightarrow \{0, 1\}$ and let $\mathcal{A}$ be a (probabilistic) algorithm that takes as input $c \in \text{Ch}^{t,w}$ and returns a string $y \in \{0, 1\}^*$. Then:

$$\sum_{i=1}^t \delta_k^V(\mathcal{A}_i, \mathcal{D}_i) \geq \frac{\varepsilon^V(\mathcal{A}) - \kappa_{t,w}}{1 - \bar{\kappa}},$$

where:

¹⁰ The verification function for $\mathcal{A}_i$ is the same $\mathbf{V}$ considered for $\mathcal{A}$, but seen as a function of the form $\text{Ch} \times (\text{Ch}^{t-1} \times \{0, 1\}^*)$.

$$\begin{aligned}
- \bar{\kappa} &= \min \left\{ \frac{w}{t} + (k-2) \frac{t-w}{t(N-1)}, (k-1) \frac{t-w}{t(N-1)} \right\}; \\
- \kappa_{t,w} &= \binom{t}{w}^{-1} \frac{\eta_{t,w}}{(N-1)^{t-w}}, \text{ with:} \\
\eta_{t,w} &= \begin{cases} \binom{w(k-1)}{w} (k-2)^{w(k-2)} (k-1)^{t-w(k-1)} & \text{if } t \geq w(k-1); \\ \binom{t}{w} (k-2)^{t-w} & \text{otherwise.} \end{cases}
\end{aligned}$$

Proof. Let $(C)_i$ be the i -th component of the random variable C uniformly distributed over $\text{Ch}^{t,w}$ and let Λ denote the event $V(C, \mathcal{A}(C)) = 1$. Therefore $\Pr[\Lambda] = \varepsilon^V(\mathcal{A})$. For $i \in \{1, \dots, t\}$, let $S_i \subset \text{Ch}$ be such that it minimizes $\delta_k^V(\mathcal{A}_i, \mathcal{D}_i)$. Then:

$$\sum_{i=1}^t \delta_k^V(\mathcal{A}_i, \mathcal{D}_i) = \sum_{i=1}^t \Pr[V(D_i, \mathcal{A}_i(D_i)) = 1 \mid D_i \notin S_i] = \sum_{i=1}^t \Pr[\Lambda \mid (C)_i \notin S_i],$$

as, for any $i \in \{1, \dots, t\}$, D_i and $(C)_i$ are identically distributed and it holds that

$$\begin{aligned}
\Pr[V(D_i, \mathcal{A}_i(D_i)) = 1 \mid D_i \notin S_i] &= \Pr[V((C)_i, \mathcal{A}_i((C)_i)) = 1 \mid (C)_i \notin S_i] \\
&= \Pr[\Lambda \mid (C)_i \notin S_i].
\end{aligned}$$

From elementary probability, it follows that:

$$\begin{aligned}
\sum_{i=1}^t \Pr[\Lambda \mid (C)_i \notin S_i] &= \sum_{i=1}^t \frac{\Pr[\Lambda \wedge (C)_i \notin S_i]}{\Pr[(C)_i \notin S_i]} = \sum_{i=1}^t \frac{\Pr[\Lambda \wedge (C)_i \notin S_i]}{1 - \Pr[(C)_i \in S_i]} \\
&\geq \frac{\Pr[\Lambda \wedge \bigcup_i (C)_i \notin S_i]}{1 - \bar{\kappa}} \geq \frac{\Pr[\Lambda] - \Pr[\bigcap_i (C)_i \in S_i]}{1 - \bar{\kappa}},
\end{aligned}$$

where in the first inequality we can take $1/(1 - \bar{\kappa})$ out of the sum by observing that, for any $i \in \{1, \dots, t\}$,

$$\Pr[(C)_i \in S_i] = \begin{cases} \frac{w}{t} + (k-2) \frac{t-w}{t(N-1)} & \text{if } 0 \in S_i; \\ (k-1) \frac{t-w}{t(N-1)} & \text{otherwise.} \end{cases}$$

Since $\bar{\kappa}$ is defined as the minimum of the previous expression, it holds that $\Pr[(C)_i \in S_i] \geq \bar{\kappa}$.

Moreover, let us define

$$\kappa_{t,w} = \max_{S_1, \dots, S_t} \Pr[(C)_1 \in S_1 \wedge (C)_2 \in S_2 \wedge \dots \wedge (C)_t \in S_t],$$

where the maximum is over all sets $S_i \subset \text{Ch}$ with $|S_i| = k-1$. Notice that, equivalently, $\kappa_{t,w} = \max_S \Pr[C \in S]$, where $S \subset \text{Ch}^{t,w}$ such that $|(S)_i| < k$ for all $i \in \{1, \dots, t\}$. The maximal size $\eta_{t,w}$ of S is computed in the following¹¹ Proposition 1 as:

$$\eta_{t,w} = \begin{cases} \binom{w(k-1)}{w} (k-2)^{w(k-2)} (k-1)^{t-w(k-1)} & \text{if } t \geq w(k-1); \\ \binom{t}{w} (k-2)^{t-w} & \text{otherwise.} \end{cases}$$

¹¹ To avoid disrupting the cryptographic narrative of the paper, the proof of this result is postponed to the end of the section.

Therefore:

$$\kappa_{t,w} = \Pr[C \in S] \leq \frac{\eta_{t,w}}{|\text{Ch}^{t,w}|} = \binom{t}{w}^{-1} \frac{\eta_{t,w}}{(N-1)^{t-w}},$$

which completes the proof. $\square$

In light of Lemma 3, we can bound the probability that at least one extractor $\mathcal{E}^{\mathcal{A}_i}(\mathcal{D}_i)$ is successful as follows:

$$\max_{1 \leq i \leq t} \delta_k^V(\mathcal{A}_i, \mathcal{D}_i) \geq \frac{1}{t} \sum_{i=1}^t \delta_k^V(\mathcal{A}_i, \mathcal{D}_i) \geq \frac{\varepsilon^V(\mathcal{A}) - \kappa_{t,w}}{t(1 - \bar{\kappa})}.$$

As a consequence, the (t, w) -fixed-weight repetition $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ of a k -special-sound Sigma protocol $(\mathcal{P}, \mathcal{V})$ is knowledge sound with knowledge error $\kappa_{t,w}$.

Theorem 1 (Fixed-Weight Repetition of a k -Special-Sound Sigma Protocol). *Let $(\mathcal{P}, \mathcal{V})$ be a k -out-of- N special-sound Sigma protocol. Let $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ be the (t, w) -fixed-weight repetition of $(\mathcal{P}, \mathcal{V})$, where $k, t \in \mathbb{N}^*$ and $1 \leq w \leq t$. Then $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ is knowledge sound with knowledge error $\kappa_{t,w}$, where:*

$$\kappa_{t,w} = \binom{t}{w}^{-1} \frac{\eta_{t,w}}{(N-1)^{t-w}},$$

with:

$$\eta_{t,w} = \begin{cases} \binom{w(k-1)}{w} (k-2)^{w(k-2)} (k-1)^{t-w(k-1)} & \text{if } t \geq w(k-1); \\ \binom{t}{w} (k-2)^{t-w} & \text{otherwise.} \end{cases}$$

Remark 4. The knowledge error $\kappa_{t,w}$ coincides with the maximum cheating probability potentially achievable by a dishonest prover against the (t, w) -fixed-weight repetition of a k -out-of- N special-sound¹² Sigma protocol $(\mathcal{P}, \mathcal{V})$.

To see this, let $\mathcal{A}$ be a dishonest prover against $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ with a fixed first message and let $S \subset \text{Ch}^{t,w}$ be the set of challenges they can correctly answer. Let C be the random variable uniformly distributed over $\text{Ch}^{t,w}$, then $\varepsilon^V(\mathcal{A}) = \Pr[C \in S] = |S|/|\text{Ch}^{t,w}|$. Notice that when $\mathcal{A}$ replies to a challenge $c \in \text{Ch}^{t,w}$, they are providing a response for t parallel executions of the underlying protocol $(\mathcal{P}, \mathcal{V})$. Since $(\mathcal{P}, \mathcal{V})$ is a k -special-sound Sigma protocol, a dishonest prover can prepare a first message for which they can correctly answer at most $k-1$ distinct challenges for the single instance of the protocol. Then, it must hold that $|(S)_i| < k$ for all $i \in \{1, \dots, t\}$. In particular, if $\eta_{t,w}$ is the maximal size of S , the maximal cheating probability of $\mathcal{A}$ is given by:

$$\varepsilon^V(\mathcal{A}) = \frac{\eta_{t,w}}{|\text{Ch}^{t,w}|}.$$

In the proof of Lemma 3, we showed that $\kappa_{t,w} = \max_S \Pr[C \in S]$, where $S \subset \text{Ch}^{t,w}$ such that $|(S)_i| < k$ for all $i \in \{1, \dots, t\}$. Since the maximal sets have the same cardinality, $\varepsilon^V(\mathcal{A})$ can match $\kappa_{t,w}$.

¹² Here we are assuming that k is minimal, i.e. $(\mathcal{P}, \mathcal{V})$ is not k' -special-sound for any $k' < k$.

Remark 5. Lately, [AFR23] considered a further generalisation, named Γ -special soundness, of the notion of k -special soundness. Within such generalisation, the subsets of challenges from which it is possible to extract a witness are determined by an arbitrary access structure Γ , i.e. a monotone set of subsets of the challenge space. The authors proved that, for any Γ -special-sound Sigma protocol, it is possible to build an extractor that has knowledge error κ_Γ and an expected running time that scales with t_Γ , where κ_Γ, t_Γ are positive integers determined by Γ . Then, if t_Γ is polynomial, Γ -special soundness implies knowledge soundness. Moreover, they showed that both a k -special-sound Sigma protocol and its t -fold parallel repetition are Γ -special sound for a suitable access structure Γ , which led them to re-discover the results of [AF22]. Similarly, the (t, w) -fixed-weight repetition of a k -special-sound 3-round interactive proof can also be described within this framework. With some additional technical steps — e.g. the combinatorial arguments developed in this paper — it is possible to obtain the same knowledge-soundness results obtained in this section by adapting the techniques from [AFR23]. However, this approach is not suitable for the multi-round case¹³. For this reason, we use the techniques from [AF22] rather than those from [AFR23], and explicitly describe the extractor for Sigma protocols. This also allows us to prove the optimality of our bounds.

We conclude this section by proving the result on the maximal size $\eta_{t,w}$ of S used within the proof of Lemma 3.

Proposition 1. *Let us consider a finite set Ch with a fixed element $\tilde{c} \in \text{Ch}$, and three positive integers k, t, w such that $|\text{Ch}| \geq k \geq 2$ and $t \geq w$. Given $S \subseteq \text{Ch}_{\tilde{c}}^{t,w}$ such that $|(S)_i| < k$ for all $i \in \{1, \dots, t\}$, we have that, if $t \geq w(k-1)$:*

$$|S| \leq \binom{w(k-1)}{w} (k-2)^{w(k-2)} (k-1)^{t-w(k-1)},$$

and otherwise:

$$|S| \leq \binom{t}{w} (k-2)^{t-w}.$$

Proof. Since $k = 2$ trivially implies $|S| \leq 1 = \binom{w}{w} 0^0 1^{t-w}$, in the remainder we suppose $k \geq 3$ ¹⁴.

For $s \in S$, let us define $Z(s)$ as the set of indices $i \in \{1, \dots, t\}$ such that the i -th entry of s is equal to $\tilde{c}$. The set $Z(S) := \bigcup_{s \in S} Z(s)$ is therefore formed by all indices i for which at least one element of S has $\tilde{c}$ as i -th entry. We denote $|Z(S)|$ by h_S , for which it holds $h_S \geq w$ by definition of $\text{Ch}_{\tilde{c}}^{t,w}$. As our goal is providing an upper bound for the cardinality of S , we can assume that, for each $i \in \{1, \dots, t\}$, it holds that $|(S)_i| = k-1$, with $\tilde{c} \in (S)_i$ if and only if $i \in Z(S)$.

¹³ In fact, it is not possible to find an access structure that suitably describes the (t, w) -fixed-weight repetition of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu+1)$ -round interactive proof.

¹⁴ Here we used $0^0 = 1$, which is a common convention in discrete mathematics, as it comes handy in many situations (e.g. the binomial theorem). For more details on this convention, see [Knu92].

Every element s of S can be thought as constructed by the following strategy. Choose a subset R of cardinality w from $Z(S)$ and, for every $i \in R$, set the i -th entry of s to $\tilde{c}$; for every $i \in Z(S) \setminus R$ choose a value in $(S)_i \setminus \{\tilde{c}\}$; finally, for every $i \in \{1, \dots, t\} \setminus Z(S)$, choose a value in $(S)_i$. This means that $|S|$ is bounded above by:

$$f(h_S) := \binom{h_S}{w} (k-2)^{h_S-w} (k-1)^{t-h_S}. \quad (1)$$

We note that $\binom{h_S}{w} (k-2)^{h_S-w}$ is monotonically increasing with ratio:

$$\frac{\binom{h_S+1}{w} (k-2)^{h_S+1-w}}{\binom{h_S}{w} (k-2)^{h_S-w}} = \frac{h_S+1}{h_S+1-w} (k-2),$$

while $(k-1)^{t-h_S}$ is monotonically decreasing with ratio:

$$\frac{(k-1)^{t-h_S}}{(k-1)^{t-(h_S+1)}} = k-1.$$

This means that f is increasing as long as:

$$\frac{h_S+1}{h_S+1-w} (k-2) > k-1 \iff h_S < w(k-1) - 1.$$

Note that:

$$\begin{aligned} f(w(k-1) - 1) &= \frac{(w(k-1) - 1)!}{w!(w(k-1) - 1 - w)!} (k-2)^{w(k-1)-1-w} (k-1)^{t-w(k-1)+1} \\ &= \frac{(w(k-1))! (w(k-1) - w)}{(w(k-1))w!(w(k-1) - w)!} \frac{(k-2)^{w(k-1)-w}}{k-2} (k-1)(k-1)^{t-w(k-1)} \\ &= \binom{w(k-1)}{w} \frac{w(k-2)}{w(k-1)} (k-2)^{w(k-1)-w} \frac{k-1}{k-2} (k-1)^{t-w(k-1)} \\ &= f(w(k-1)), \end{aligned}$$

so, in conclusion, if $t \geq w(k-1)$ we have:

$$|S| \leq f(w(k-1)) = \binom{w(k-1)}{w} (k-2)^{w(k-2)} (k-1)^{t-w(k-1)}.$$

On the other hand, if $t < w(k-1)$, then f is increasing up to t , so:

$$|S| \leq f(t) = \binom{t}{w} (k-2)^{t-w}. \quad \square$$

4 Combinatorial Bounds

Our next goal is to prove that a fixed-weight repetition of a multi-round special-sound interactive proof still enjoys knowledge soundness. To do this, we will

employ some combinatorial bounds. The concepts and results relative to such bounds are much more involved compared to the 3-round case (see Proposition 1), and so we devote this section entirely to them.

The bounds we will present may be of independent interest, and for this reason we will state and prove them in full generality. Nevertheless, as they will find a natural application in Sections 3 and 5, we will try to use the same notation that will be used there.

In the following lines, we provide a high-level overview of the section. By reading this introduction, a reader not interested in engaging with every formal step necessary to rigorously prove the combinatorial results can have a summary of the most significant outcomes.

The main problem addressed in this section arises from the need to estimate, for the fixed-weight repetition of a given multi-round special-sound interactive proof, the maximum success probability of a dishonest prover. This is derived from the maximum cardinality of the sets of transcripts/challenges from which it is not possible to extract a witness.

Therefore, we start by giving a formal definition of the sets of transcripts/challenges from which it is not possible to extract a witness. For convenience, we refer to such sets as *acceptable sets* (Definition 17). To simplify the computation, we resort to Lemma 4, which skims these sets, and reduce the considered problem to finding the maximum over specific classes of acceptable sets, which we call *saturated acceptable sets*. Following Definition 21, we can characterize each saturated acceptable set via a sequence $(d_{i,b_i})_{i \in \{1, \dots, t\}}$ that basically represents the cardinality of some specific subsets. More precisely, Lemma 5 shows that the cardinality of a saturated acceptable set can be expressed by a formula which depends on its sequence $(d_{i,b_i})_{i \in \{1, \dots, t\}}$, and that two other important properties are satisfied by $(d_{i,b_i})_{i \in \{1, \dots, t\}}$ (Equation (2) and Equation (3)). Then, by relying on a technical lemma on sequences of non-negative integers (Lemma 6), we prove in Proposition 2 an upper bound for the cardinalities of acceptable sets.

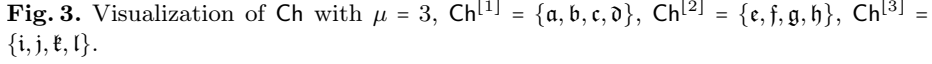
4.1 Bound for the $(2\mu + 1)$ -round case

In the remainder of this section $k_1, \dots, k_\mu, N_1, \dots, N_\mu$ will denote positive integers while $\text{Ch}^{[1]}, \dots, \text{Ch}^{[\mu]}$ are finite sets such that $|\text{Ch}^{[\ell]}| = N_\ell \geq k_\ell \geq 2 \ \forall \ell \in \{1, \dots, \mu\}$. We set $\text{Ch} = \prod_{\ell=1}^\mu \text{Ch}^{[\ell]}$ and we fix an element $\tilde{c}$ in $\text{Ch}^{[\mu]}$.

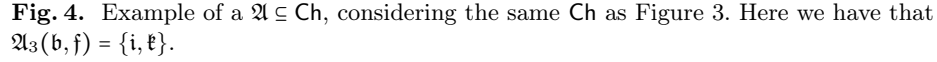
As exemplified by Figure 3, we can visualize the sequences of Ch as a tree whose root is the empty sequence, and given a node at the level $\ell - 1$ for $\ell \in \{1, \dots, \mu\}$, its childs are all the elements of $\text{Ch}^{[\ell]}$. That is, we can see the sequences as paths on this tree from the root to a leaf, and a set of sequences can be seen as a subtree.

Definition 15. For any $S \subseteq \text{Ch}$, $\ell \in \{2, \dots, \mu\}$ and $(s_1, \dots, s_{\ell-1}) \in \prod_{j=1}^{\ell-1} \text{Ch}^{[j]}$, we define:

$$S_\ell(s_1, \dots, s_{\ell-1}) := \{s \in \text{Ch}^{[\ell]} : \exists (s_1, \dots, s_{\ell-1}, s, r_{\ell+1}, \dots, r_\mu) \in S\}.$$



Informally, $S_\ell(s_1, \dots, s_{\ell-1})$ denotes the set of ℓ -th entries of the elements of S with the first $\ell - 1$ entries equal to $(s_1, \dots, s_{\ell-1})$ (see Figure 4 for an example).



Definition 16 (Acceptability Predicate). Given $S \subseteq \text{Ch}$, for any $(a_1, \dots, a_\mu) \in \text{Ch}$ we define the predicate $P_{S,\mu}$ as follows:

$$P_{S,\mu}((a_1, \dots, a_\mu)) \iff (a_1, \dots, a_\mu) \in S.$$

For $\ell \in \{1, \dots, \mu - 1\}$, the predicate $P_{S,\ell}((a_1, \dots, a_\ell))$ is true if and only if:

$$|\{(a_1, \dots, a_\ell, a_{\ell+1}) : a_{\ell+1} \in S_{\ell+1}(a_1, \dots, a_\ell) \wedge P_{S, \ell+1}(a_1, \dots, a_{\ell+1})\}| \geq k_{\ell+1}.$$

Example 1. Considering the set $\mathfrak{A}$ of Figure 4, and fixing $k_1 = 3, k_2 = 2, k_3 = 3$; we have that $P_{\mathfrak{A},3}((\mathfrak{d}, \mathfrak{h}, \mathfrak{i}))$ is true, $P_{\mathfrak{A},2}((\mathfrak{b}, \mathfrak{g}))$ is false, $P_{\mathfrak{A},1}((\mathfrak{c}))$ is true.

We aim at finding an upper bound of the cardinality of particular sets $S \subseteq \text{Ch}_c^{t,w}$ that do not define a $(k_1, \dots, k_\mu)$ -tree of accepting transcripts in the context of Definition 8. We call these sets *acceptable sets*. The mentioned bound determines the maximum cheating probability of a dishonest prover in the case of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu+1)$ -round interactive proof for which we consider a (t, w) -fixed-weight parallel repetition.

Definition 17 (Acceptable Set). Given $A \subseteq \text{Ch}_c^{t,w}$, we say that it is a $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ acceptable set (or, when clear from the context, simply acceptable set) if the following condition holds:

$$|\{a_1 : a_1 \in (A)_i^{[1]} \wedge P_{(A)_i,1}(a_1)\}| < k_1 \quad \forall i \in \{1, \dots, t\}.$$

Example 2. Building on Example 1, for $t = 1$ we have that $\mathfrak{A}$ is a $(3, 2, 3)$ -out-of- $(4, 4, 4)$ acceptable set.

In this work a subfamily of acceptable sets are of particular importance. We name them *saturated acceptable sets*. Before formally defining them, we provide some introductive definitions.

First we define the concept of *saturating choice*. Intuitively, we want to build a maximal acceptable set by adding as many elements as possible without breaking acceptability. Recalling the tree visualization, note that the acceptability predicate tells that, at each level $\ell \in \{1, \dots, \mu - 1\}$, given a path $(a_1, \dots, a_{\ell-1})$ up to $k_\ell - 1$ children a_ℓ of $a_{\ell-1}$ may be such that $P_{S,\ell}((a_1, \dots, a_{\ell-1}, a_\ell))$ is true and still have $P_{S,\ell-1}((a_1, \dots, a_{\ell-1}))$ false. This means that we can add to the set the complete sub-tree rooted in those a_ℓ without breaking acceptability. Indeed, the saturating choice formalizes a way of choosing complete sub-trees to add to an acceptable set without breaking acceptability, by defining a set $\mathcal{S}$ of sequences that represent the nodes which are not the root of a complete sub-tree, and a map $\bar{B}$ that, for each of these nodes, tells which children are roots of complete sub-trees.

Example 3. Building again on Example 1, we have that

- $\mathcal{S} = \{\emptyset, (\mathfrak{b}), (\mathfrak{d}), (\mathfrak{b}, \mathfrak{f}), (\mathfrak{b}, \mathfrak{g}), (\mathfrak{b}, \mathfrak{h}), (\mathfrak{d}, \mathfrak{c}), (\mathfrak{d}, \mathfrak{f}), (\mathfrak{d}, \mathfrak{h})\}$;
- $\bar{B}(\emptyset) = \{\mathfrak{a}, \mathfrak{c}\}$, $\bar{B}((\mathfrak{b})) = \{\mathfrak{e}\}$, $\bar{B}((\mathfrak{d})) = \{\mathfrak{g}\}$, $\bar{B}((\mathfrak{b}, \mathfrak{f})) = \{\mathfrak{i}, \mathfrak{k}\}$, $\bar{B}((\mathfrak{b}, \mathfrak{g})) = \{\mathfrak{j}, \mathfrak{l}\}$, $\bar{B}((\mathfrak{b}, \mathfrak{h})) = \{\mathfrak{k}, \mathfrak{l}\}$, $\bar{B}((\mathfrak{d}, \mathfrak{c})) = \{\mathfrak{j}, \mathfrak{l}\}$, $\bar{B}((\mathfrak{d}, \mathfrak{f})) = \{\mathfrak{i}, \mathfrak{l}\}$, $\bar{B}((\mathfrak{d}, \mathfrak{h})) = \{\mathfrak{i}, \mathfrak{j}\}$.

Definition 18 (Saturating Choice). Given μ finite sets $\text{Ch}^{[1]}, \dots, \text{Ch}^{[\mu]}$, and $\forall \ell \in \{1, \dots, \mu\}$, $k_\ell \leq |\text{Ch}^{[\ell]}|$, a saturating choice is a pair $(\mathcal{S}, \bar{B})$ such that:

- $\mathcal{S} \subseteq \bigcup_{\ell=0}^{\mu-1} \prod_{j=1}^{\ell} \text{Ch}^{[j]}$ is a set of sequences;
- $\bar{B} : \mathcal{S} \rightarrow \bigcup_{\ell=1}^{\mu} \mathcal{P}(\text{Ch}^{[\ell]})$ is a function such that for every $s \in \mathcal{S}$ with $|s| = \ell$, associates to s a set $\bar{B}(s) \subseteq \text{Ch}^{[\ell+1]}$, where $\mathcal{P}(\text{Ch}^{[\ell]})$ denotes the power set of $\text{Ch}^{[\ell]}$.

Furthermore, the pair $(\mathcal{S}, \bar{B})$ satisfies the following properties:

- $\emptyset \in \mathcal{S}$, where $\emptyset$ denotes the empty sequence;
- $\bar{B}(\emptyset) \subseteq \text{Ch}^{[1]}$ is such that $|\bar{B}(\emptyset)| = k_1 - 1$;
- $\forall a_1 \in \text{Ch}^{[1]} \setminus \bar{B}(\emptyset)$, $(a_1) \in \mathcal{S}$;
- $\forall (a_1) \in \mathcal{S}$, $\bar{B}((a_1)) \subseteq \text{Ch}^{[2]}$ is a set such that $|\bar{B}((a_1))| = k_2 - 1$;

and, inductively for $\ell \in \{2, \dots, \mu - 1\}$:

- $\forall (a_1, \dots, a_\ell) \in \prod_{j=1}^{\ell} (\text{Ch}^{[j]} \setminus \bar{B}((a_1, \dots, a_{j-1})))$, $(a_1, \dots, a_\ell) \in \mathcal{S}$;
- $\forall (a_1, \dots, a_\ell) \in \mathcal{S}$, $\bar{B}((a_1, \dots, a_\ell)) \subseteq \text{Ch}^{[\ell+1]}$ satisfies $|\bar{B}((a_1, \dots, a_\ell))| = k_{\ell+1} - 1$.

The next definition formalizes the operator $\text{sat}()$ that, given a sequence (i.e., a node in the tree visualization) and a set of possible next elements (i.e., children of the node), returns the set of all possible sequences that have that prefix and continue with an element of that set (i.e. all the leaves of the complete sub-trees rooted in these children).

Definition 19. Given μ finite sets $\text{Ch}^{[1]}, \dots, \text{Ch}^{[\mu]}$, $\ell < \mu$, $k_{\ell+1} \leq |\text{Ch}^{[\ell+1]}|$, a sequence $(a_1, \dots, a_\ell) \in \prod_{j=1}^\ell \text{Ch}^{[j]}$, and a set $\bar{B} \subseteq \text{Ch}^{[\ell+1]}$ such that $|\bar{B}| = k_{\ell+1} - 1$, we can define the set:

$$\text{sat}((a_1, \dots, a_\ell), \bar{B}) := \left\{ (a_1, \dots, a_\ell, a_{\ell+1}, a_{\ell+2}, \dots, a_\mu) : \right. \\ \left. a_{\ell+1} \in \bar{B} \wedge (a_{\ell+2}, \dots, a_\mu) \in \prod_{j=\ell+2}^\mu \text{Ch}^{[j]} \right\}.$$

Definition 20 (Saturated Acceptable Set). Given $\bar{A} \subseteq \text{Ch}_{\tilde{c}}^{t,w}$, we say that it is a $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ saturated acceptable set (or, when clear from the context, simply saturated acceptable set) if $\forall i \in \{1, \dots, t\}$, there is a saturating choice $(\mathcal{S}_i, \bar{B}_i)$ such that $(\bar{A})_i = \bigsqcup_{s \in \mathcal{S}_i} \text{sat}(s, \bar{B}_i(s))$ (see Definitions 18 and 19). Moreover we have that $\forall ((y)_1, \dots, (y)_t) \in \prod_{i=1}^t (\bar{A})_i \cap \text{Ch}_{\tilde{c}}^{t,w}$, $((y)_1, \dots, (y)_t) \in \bar{A}$.

Example 4. Building on Example 3, for $t = 1$ (and ignoring $\tilde{c}$), we have that $\mathfrak{A}$ is a saturated acceptable set, precisely with the saturating choice seen above.

Note that sets that satisfy Definition 20 are indeed acceptable sets because, by construction, we have that:

$$|\{a_1 : a_1 \in (A)_i^{[1]} \wedge P_{(A)_i,1}(a_1)\}| = k_1 - 1, \quad \forall i \in \{1, \dots, t\}.$$

In fact, the saturating choices $(\mathcal{S}_i, \bar{B}_i)$ have the property that, $\forall \ell \in \{1, \dots, \mu\}$:

$$|\{s \in \mathcal{S}_i : (s, a_{\ell+1}, \dots, a_\mu) \in (A)_i \wedge P_{(A)_i,\ell}(s)\}| = k_\ell - 1.$$

This gives the intuition that there is no margin left to add more elements to saturated acceptable sets while retaining their acceptability. More formally, the following lemma proves that every maximal acceptable set is a saturated acceptable set.

Lemma 4 (Maximality of Saturated Acceptable Sets). For every $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ acceptable set $A \subseteq \text{Ch}_{\tilde{c}}^{t,w}$ there exists a saturated acceptable set $\bar{A} \subseteq \text{Ch}_{\tilde{c}}^{t,w}$ such that $\bar{A} \supseteq A$.

Proof. Let $A \subseteq \text{Ch}_{\tilde{c}}^{t,w}$ be a $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ acceptable set. To construct a saturated acceptable set $\bar{A} \subseteq \text{Ch}_{\tilde{c}}^{t,w}$ which contains A , we will build an appropriate saturating choice (see Definition 18) $(\mathcal{S}_i, \bar{B}_i)$ for all $i \in t$. We initially set $\mathcal{S}_i := \{\emptyset\}$ and then iteratively add elements to it.

Let us define $B_i(\odot) := \{a_1 : a_1 \in (A)_i^{[1]} \wedge P_{(A)_i,1}(a_1)\}$. By definition of acceptable set, we have that $|B_i(\odot)| \leq k_1 - 1$, so we can define $\bar{B}_i(\odot) \subseteq \text{Ch}^{[1]}$ to be a set such that $B_i(\odot) \subseteq \bar{B}_i(\odot)$ and $|\bar{B}_i(\odot)| = k_1 - 1$.

For each $a_1 \in \text{Ch}^{[1]} \setminus \bar{B}_i(\odot)$, we add (a_1) to $\mathcal{S}_i$, and we define the set $B_i((a_1)) := \{a_2 : a_2 \in (A)_i^{[2]} \wedge P_{(A)_i,2}((a_1, a_2))\}$. Again, by definition of acceptable set, we have that $|B_i((a_1))| \leq k_2 - 1$, so we can define $\bar{B}_i((a_1)) \subseteq \text{Ch}^{[2]}$ to be a set such that $B_i((a_1)) \subseteq \bar{B}_i((a_1))$ and $|\bar{B}_i((a_1))| = k_2 - 1$.

In general, for $\ell \in \{2, \dots, \mu - 1\}$ and for any $a_\ell \in \text{Ch}^{[\ell]} \setminus \bar{B}_i((a_1, \dots, a_{\ell-1}))$, we add $(a_1, \dots, a_\ell)$ to $\mathcal{S}_i$ and define:

$$B_i((a_1, \dots, a_\ell)) := \{a_{\ell+1} : a_{\ell+1} \in (A)_i^{[\ell+1]} \wedge P_{(A)_i, \ell+1}((a_1, \dots, a_\ell, a_{\ell+1}))\}.$$

As before, we have that $|B_i((a_1, \dots, a_\ell))| \leq k_{\ell+1} - 1$, so it is possible to define $\bar{B}_i((a_1, \dots, a_\ell)) \subseteq \text{Ch}^{[\ell+1]}$ as a set such that:

$$B_i((a_1, \dots, a_\ell)) \subseteq \bar{B}_i((a_1, \dots, a_\ell)) \wedge |\bar{B}_i((a_1, \dots, a_\ell))| = k_{\ell+1} - 1.$$

By construction, the saturated acceptable set $\bar{A}$ associated to these saturating choices $(\mathcal{S}_i, \bar{B}_i)_{i \in \{1, \dots, t\}}$ contains $(A)_i$. $\square$

Definition 21. Given a saturated acceptable set $\bar{A} \subseteq \text{Ch}_c^{t,w}$, it is possible to associate to $\bar{A}$ a set of t -sequences $\left\{ (d(\bar{A})_{1,b_1}, \dots, d(\bar{A})_{t,b_t}) : (b_1, \dots, b_t) \in (\text{Ch}^{[\mu]})_c^{t,w} \right\}$, where we define:

$$d(\bar{A})_{i,b_i} := \left| \left\{ (a)_i \in (\bar{A})_i : (a)_i^{[\mu]} = b_i \right\} \right| \quad \forall i \in \{1, \dots, t\}.$$

The result below provides a first characterization of the cardinality of a saturated acceptable set, exploiting the t -sequences defined above.

Lemma 5. Let $\bar{A} \subseteq \text{Ch}_c^{t,w}$ be a $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ saturated acceptable set. Then, for every $i \in \{1, \dots, t\}$:

$$\sum_{x \in \text{Ch}^{[\mu]}} d(\bar{A})_{i,x} = \sum_{\ell=1}^{\mu} \left(\prod_{j=\ell+1}^{\mu} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right); \quad (2)$$

$$d(\bar{A})_{i,\bar{c}} \geq \sum_{\ell=1}^{\mu-1} \left(\prod_{j=\ell+1}^{\mu-1} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right); \quad (3)$$

$$\text{and } |\bar{A}| = \sum_{b \in (\text{Ch}^{[\mu]})_c^{t,w}} \prod_{i=1}^t d(\bar{A})_{i,(b)_i}.$$

Proof. Let $\bar{A} \subseteq \text{Ch}_c^{t,w}$ be a $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ saturated acceptable set. By Definitions 18 and 19, we have that, for all $i \in \{1, \dots, t\}$, $\ell \in \{0, \dots, \mu - 1\}$, and $s \in \mathcal{S}_i$ such that $|s| = \ell$:

$$|\text{sat}(s, \bar{B}_i(s))| = (k_{\ell+1} - 1) \cdot \prod_{j=\ell+2}^{\mu} N_j.$$

Moreover, note that for $\ell > 0$ there are $N_\ell - k_\ell + 1$ choices for the last element of s if we fix the first $\ell - 1$, so in total we have that:

$$|\{s \in \mathcal{S}_i : |s| = \ell\}| = \prod_{j=1}^{\ell} (N_j - k_j + 1).$$

Putting everything together, setting to 1 the empty product, we have that:

$$|\bigsqcup_{s \in \mathcal{S}_i : |s| = \ell} \text{sat}(s, \bar{B}_i(s))| = \left(\prod_{j=1}^{\ell} (N_j - k_j + 1) \right) (k_{\ell+1} - 1) \left(\prod_{j=\ell+2}^{\mu} N_j \right),$$

so, by Definition 20 we have the following relation:

$$|(\bar{A})_i| = \sum_{\ell=1}^{\mu} \left(\prod_{j=\ell+1}^{\mu} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right). \quad (4)$$

Thanks to the second condition of Definition 20, we have that:

$$(\bar{A})_i = \bigsqcup_{x \in \text{Ch}^{[\mu]}} \left\{ (a)_i \in (\bar{A})_i : (a)_i^{[\mu]} = x \right\},$$

as for every $(y)_i \in (\bar{A})_i$ there exist $(y)_1, \dots, (y)_{i-1}, (y)_i, (y)_{i+1}, \dots, (y)_t$ such that $((y)_1, \dots, (y)_t) \in \bar{A}$. So by Definition 21 we have proved Equation (2). Another direct consequence of this is that $|\bar{A}| = \sum_{b \in \{\text{Ch}^{[\mu]}\}_{\bar{c}}^{t,w}} \prod_{i=1}^t d(\bar{A})_{i,(b)_i}$.

Finally, for any $i \in \{1, \dots, t\}$ we have that:

$$d(\bar{A})_{i,\bar{c}} \geq \sum_{\ell=1}^{\mu-1} \left(\prod_{j=\ell+1}^{\mu-1} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right),$$

as the right-hand side corresponds to the number of saturated branches in $(\bar{A})_i$, i.e. the number of elements $((y)_i^{[1]}, \dots, (y)_i^{[\mu-1]}) \in \prod_{i=1}^{\mu-1} \text{Ch}^{[i]}$ such that, for every x in $\text{Ch}^{[\mu]}$, the element $((y)_i^{[1]}, \dots, (y)_i^{[\mu-1]}, x)$ belongs to $(\bar{A})_i$. $\square$

The result we present below is a technical lemma regarding the property of a particular sequence of sequences. It will be a key step in proving the maximum cardinality among the acceptable sets.

Lemma 6. *Let $w \leq t$, N, Z_0, Z_1, Z_2 be positive integers and consider t sequences of non-negative integers $(d_{i,0}, \dots, d_{i,N})_{i \in \{1, \dots, t\}}$ such that, for every $i \in \{1, \dots, t\}$, it holds that:*

$$d_{i,0} \in \{Z_0, Z_2\}, \quad \sum_{j=0}^N d_{i,j} = Z_1.$$

Let $\alpha := |\{i : d_{i,0} = Z_0\}|$, $\ell_0 := \max(0, w - t + \alpha)$, and $\ell_1 := \min(w, \alpha)$. Then:

$$\sum_{b \in S_{t,w}} \prod_{i=1}^t d_{i,b_i} = \sum_{\ell=\ell_0}^{\ell_1} \binom{\alpha}{\ell} \binom{t-\alpha}{w-\ell} Z_0^\ell (Z_1 - Z_0)^{\alpha-\ell} (Z_2)^{w-\ell} (Z_1 - Z_2)^{t-\alpha-w+\ell}, \quad (5)$$

where $S_{t,w} := \{0, \dots, N\}_0^{t,w}$ (see Definition 11).

Proof. Given $b \in S_{t,w}$, let us define four support sets:

$$\begin{aligned} B_{1,b} &:= \{i \in \{1, \dots, t\} : b_i = 0 \wedge d_{i,0} = Z_0\}, \\ B_{2,b} &:= \{i \in \{1, \dots, t\} : b_i = 0 \wedge d_{i,0} = Z_2\}, \\ B_{3,b} &:= \{i \in \{1, \dots, t\} : b_i \neq 0 \wedge d_{i,0} = Z_0\}, \\ B_{4,b} &:= \{i \in \{1, \dots, t\} : b_i \neq 0 \wedge d_{i,0} = Z_2\}. \end{aligned}$$

It clearly holds that $t = |\sqcup_{j=1}^4 B_{j,b}|$, $\alpha = |B_{1,b} \sqcup B_{3,b}|$, $w = |B_{1,b} \sqcup B_{2,b}|$. Therefore, we have:

$$\begin{aligned} \sum_{b \in S_{t,w}} \prod_{i=1}^t d_{i,b_i} &= \sum_{b \in S_{t,w}} \prod_{j=1}^4 \left(\prod_{i \in B_{j,b}} d_{i,b_i} \right) \\ &= \sum_{b \in S_{t,w}} \prod_{i \in B_{1,b}} d_{i,0} \prod_{i \in B_{2,b}} d_{i,0} \prod_{i \in B_{3,b}} d_{i,b_i} \prod_{i \in B_{4,b}} d_{i,b_i} \\ &= \sum_{b \in S_{t,w}} \prod_{i \in B_{1,b}} Z_0 \prod_{i \in B_{2,b}} Z_2 \prod_{i \in B_{3,b}} d_{i,b_i} \prod_{i \in B_{4,b}} d_{i,b_i}. \end{aligned}$$

Now let us consider two disjunct sets $B_1, B_2 \subseteq \{1, \dots, t\}$, with $\ell := |B_1|$. In order for the set $B' := \{b \in S_{t,w} : B_{1,b} = B_1 \wedge B_{2,b} = B_2\}$ to be non-empty, we have that $\ell \leq \min(w, \alpha)$, and also $|B_2| = w - \ell \leq t - \alpha$, i.e., $\ell \geq \max(0, w - t + \alpha)$. Now, we have that:

$$\begin{aligned} \sum_{b \in B'} \prod_{i=1}^t d_{i,b_i} &= \sum_{b \in B'} \prod_{i \in B_1,b} Z_0 \prod_{i \in B_2,b} Z_2 \prod_{i \in B_3,b} d_{i,b_i} \prod_{i \in B_4,b} d_{i,b_i} \\ &= \sum_{b \in B'} (Z_0)^\ell (Z_2)^{w-\ell} \prod_{i \in B_3,b} d_{i,b_i} \prod_{i \in B_4,b} d_{i,b_i}. \end{aligned}$$

It is straightforward to see that, for any $i' \in \{1, \dots, t\}$, the equality below holds:

$$\begin{aligned} S_{t,w} &= \left(\bigsqcup_{b' \in S_{t-1,w}} \bigsqcup_{j=1}^N \{(b'_1, \dots, b'_{i'-1}, j, b'_{i'}, \dots, b'_{t-1})\} \right) \\ &\quad \bigsqcup \\ &\quad \left(\bigsqcup_{b' \in S_{t-1,w-1}} \{(b'_1, \dots, b'_{i'-1}, 0, b'_{i'}, \dots, b'_{t-1})\} \right). \end{aligned}$$

Now we want to adapt this partition to our set $B' \subseteq S_{t,w}$ by considering an index i' in $\{1, \dots, t\} \setminus (B_1 \sqcup B_2)$.

Let us define an index-translation function $f_{i'} : \{1, \dots, t-1\} \longrightarrow \{1, \dots, t\} \setminus \{i'\}$:

$$f_{i'}(i) := \begin{cases} i & \text{if } i < i'; \\ i+1 & \text{if } i \geq i'. \end{cases}$$

Then, we can define the following subsets of $\{1, \dots, t-1\}$:

$$B_1(i') := \{i : f_{i'}(i) \in B_1\}, \quad B_2(i') := \{i : f_{i'}(i) \in B_2\},$$

and introduce the set:

$$B'(i') := \{b \in S_{t-1,w} : \{i \in \{1, \dots, t-1\} : b_i = 0 \wedge d_{f_{i'}(i),0} = Z_0\} = B_1(i') \wedge \\ \wedge \{i \in \{1, \dots, t-1\} : b_i = 0 \wedge d_{f_{i'}(i),0} = Z_2\} = B_2(i')\}.$$

We have that:

$$B' = \bigsqcup_{b' \in B'(i')} \bigsqcup_{j=1}^N \{(b'_1, \dots, b'_{i'-1}, j, b'_{i'}, \dots, b'_{t-1})\}.$$

Let us define also:

$$B_3(i') := \{i \in \{1, \dots, t-1\} : f_{i'}(i) \notin B_1 \sqcup B_2 \wedge d_{f_{i'}(i),0} = Z_0\}; \\ B_4(i') := \{i \in \{1, \dots, t-1\} : f_{i'}(i) \notin B_1 \sqcup B_2 \wedge d_{f_{i'}(i),0} = Z_2\}.$$

We can use this partition in our sum. We first assume $d_{i',0} = Z_0$, which leads to:

$$\begin{aligned} \sum_{b \in B'} \prod_{i=1}^t d_{i,b_i} &= (Z_0)^\ell (Z_2)^{w-\ell} \sum_{b \in B'} \left(\prod_{i \in B_3,b} d_{i,b_i} \prod_{i \in B_4,b} d_{i,b_i} \right) \\ &= (Z_0)^\ell (Z_2)^{w-\ell} \sum_{b' \in B'(i')} \sum_{j=1}^N \left(d_{i',j} \prod_{i \in B_3(i')} d_{f_{i'}(i),b'_i} \prod_{i \in B_4(i')} d_{f_{i'}(i),b'_i} \right) \\ &= (Z_0)^\ell (Z_2)^{w-\ell} \sum_{b' \in B'(i')} \left(\sum_{j=1}^N d_{i',j} \right) \left(\prod_{i \in B_3(i')} d_{f_{i'}(i),b'_i} \prod_{i \in B_4(i')} d_{f_{i'}(i),b'_i} \right) \\ &= (Z_0)^\ell (Z_2)^{w-\ell} \sum_{b' \in B'(i')} \left(\left(\sum_{j=0}^N d_{i',j} \right) - d_{i',0} \right) \left(\prod_{i \in B_3(i')} d_{f_{i'}(i),b'_i} \prod_{i \in B_4(i')} d_{f_{i'}(i),b'_i} \right) \\ &= (Z_0)^\ell (Z_2)^{w-\ell} \sum_{b' \in B'(i')} (Z_1 - Z_0) \left(\prod_{i \in B_3(i')} d_{f_{i'}(i),b'_i} \prod_{i \in B_4(i')} d_{f_{i'}(i),b'_i} \right) \\ &= (Z_0)^\ell (Z_2)^{w-\ell} (Z_1 - Z_0) \sum_{b' \in B'(i')} \left(\prod_{i \in B_3(i')} d_{f_{i'}(i),b'_i} \prod_{i \in B_4(i')} d_{f_{i'}(i),b'_i} \right). \end{aligned}$$

Note that in this way we have effectively extracted the factor corresponding to the index i' without affecting the other indices, and note also that in the case where $d_{i',0} = Z_2$, the only difference is that we can factor out $(Z_1 - Z_2)$ instead of $(Z_1 - Z_0)$. If we repeat this technique starting from $B'(i')$ (whose elements have length $t-1$) instead of B' (whose elements have length t), we can factor out another index. This means that, by repeating the same partition and factoring technique for every $i' \in \{1, \dots, t\} \setminus (B_1 \sqcup B_2)$, and remembering that for any $b \in B'$ we have $|B_{3,b}| = \alpha - \ell$ and $|B_{4,b}| = t - \alpha - w + \ell$, we obtain:

$$\sum_{b \in B'} \prod_{i=1}^t d_{i,b_i} = (Z_0)^\ell (Z_2)^{w-\ell} (Z_1 - Z_0)^{\alpha-\ell} (Z_1 - Z_2)^{t-\alpha-w+\ell}.$$

To conclude, note that for ℓ fixed there are $\binom{\alpha}{\ell}$ possible choices for B_1 and $\binom{t-\alpha}{w-\ell}$ possible choices for B_2 , and that by varying ℓ , the set of all the possible B' forms a partition of $S_{t,w}$. $\square$

The following result provides an upper bound for the cardinalities of (saturated) acceptable sets.

Proposition 2. *Denote by $n_{t,w}$ the maximum cardinality of a $(k_1, \dots, k_\mu)$ -out-of- $(N_1, \dots, N_\mu)$ acceptable set. Then, $n_{t,w}$ is the maximum of the expression:*

$$\sum_{\ell=\max(0, w-t+\alpha)}^{\min(w, \alpha)} \binom{\alpha}{\ell} \binom{t-\alpha}{w-\ell} Z_0^\ell (Z_1 - Z_0)^{\alpha-\ell} (Z_2)^{w-\ell} (Z_1 - Z_2)^{t-\alpha-w+\ell},$$

where the maximum is taken over $\alpha \in \{0, \dots, t\}$ and:

$$Z_0 := \prod_{\ell=1}^{\mu-1} N_\ell; \quad (6)$$

$$Z_1 := \sum_{\ell=1}^{\mu} \left(\prod_{j=\ell+1}^{\mu} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right); \quad (7)$$

$$Z_2 := \sum_{\ell=1}^{\mu-1} \left(\prod_{j=\ell+1}^{\mu-1} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right). \quad (8)$$

Proof. Notice that, thanks to Lemma 4, we can always limit ourselves to consider saturated acceptable sets $\bar{A}$ for which, thanks to Lemma 5, the following conditions hold for every $i \in \{1, \dots, t\}$:

$$\sum_{x \in \text{Ch}^{[\mu]}} d(\bar{A})_{i,x} = Z_1, \quad d(\bar{A})_{i,\tilde{c}} \geq Z_2, \quad (9)$$

with Z_1, Z_2 as in Eq. 7 and 8, respectively. In addition,

$$n_{t,w} := \max_A \{|A|\} = \max_{\bar{A}} \{|\bar{A}|\} = \max_{\bar{A}} \left\{ \sum_{b \in (\text{Ch}^{[\mu]})_{\tilde{c}}^{t,w}} \prod_{i=1}^t d(\bar{A})_{i,(b)_i} \right\}. \quad (10)$$

For simplicity of notation we remove below the dependence from $\bar{A}$, meaning that we replace $d(\bar{A})_{i,x}$ with $d_{i,x}$, with $x \in \{0, \dots, N_\mu - 1\}$. Furthermore, we biject the elements of $\text{Ch}^{[\mu]}$ with the set $\{0, \dots, N_\mu - 1\}$ mapping $\tilde{c}$ to 0, and again we define $S_{t,w} := \{0, \dots, N_\mu - 1\}_0^{t,w}$. According to this, we will also denote by $\{(\bar{d}_{i,0}, \dots, \bar{d}_{i,N_\mu-1})\}_{i \in \{1, \dots, t\}}$ a generic set that respects the conditions imposed by Equation (9). We would therefore like to find a valid assignment of $(\bar{d}_{i,j})_{i,j}$ that maximizes the expression above. We show that it is sufficient to consider sets such that, for each index $i \in \{1, \dots, t\}$:

$$\left(\bar{d}_{i,0}, \sum_{j=1}^{N_\mu-1} \bar{d}_{i,j} \right) \in \{(Z_0, Z_1 - Z_0), (Z_2, Z_1 - Z_2)\}. \quad (11)$$

5. FIXED-WEIGHT REPETITION OF A $(k_1, \dots, k_\mu)$ -SPECIAL-SOUND INTERACTIVE PROOF

Consider the case $i = 1$ (the extension to the generic case $i \neq 1$ is immediate) and notice that:

$$n_{t,w} = \max \left\{ \bar{d}_{1,0} \sum_{(b_2, \dots, b_t) \in S_{t-1, w-1}} \prod_{i=2}^t \bar{d}_{i, b_i} + \left(\sum_{j=1}^{N_\mu-1} \bar{d}_{1,j} \right) \sum_{(b_2, \dots, b_t) \in S_{t-1, w}} \prod_{i=2}^t \bar{d}_{i, b_i} \right\}.$$

Let us consider the two sums in the expression above, namely:

$$\sum_{(b_2, \dots, b_t) \in S_{t-1, w-1}} \prod_{i=2}^t \bar{d}_{i, b_i} \quad \text{and} \quad \sum_{(b_2, \dots, b_t) \in S_{t-1, w}} \prod_{i=2}^t \bar{d}_{i, b_i}.$$

Depending on whether the first sum is larger or smaller than the second, we can maximize the entire expression by maximizing the factor that multiplies the largest sum. According to this, we can maximize $n_{t,w}$ by maximizing either $\bar{d}_{1,0}$ or $\sum_{j=1}^{N_\mu-1} \bar{d}_{1,j}$. In particular, if we have that:

$$\sum_{(b_2, \dots, b_t) \in S_{t-1, w-1}} \prod_{i=2}^t \bar{d}_{i, b_i} \geq \sum_{(b_2, \dots, b_t) \in S_{t-1, w}} \prod_{i=2}^t \bar{d}_{i, b_i},$$

then the maximum is obtained when we take $\bar{d}_{1,0} = \prod_{\ell=1}^{\mu-1} N_\ell = Z_0$ (and consequently $\sum_{j=1}^{N_\mu-1} \bar{d}_{1,j} = Z_1 - Z_0$). Otherwise, since $\bar{d}_{1,0}$ is always greater or equal than Z_2 , we can maximize the expression by choosing $\sum_{j=1}^{N_\mu-1} \bar{d}_{1,j} = Z_1 - Z_2$ and $\bar{d}_{1,0} = Z_2$.

Since we can consider only sets of this type when maximizing Eq. 10, Lemma 6 applies with Z_0, Z_1 and Z_2 respectively as in Eq. 6, 7 and 8, $N := N_\mu - 1$. This concludes the proof. $\square$

5 Fixed-Weight Repetition of a $(k_1, \dots, k_\mu)$ -Special-Sound Interactive Proof

In the following, we extend the result of Section 3 to multi-round interactive proofs. Let $(\mathcal{P}, \mathcal{V})$ be a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proof with challenge space $\text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$. We define $K = \prod_{i=1}^\mu k_i$ and write $\mathbf{c} = (c^{[1]}, \dots, c^{[\mu]})$ for an element $\mathbf{c} \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$.

Similarly to the case of 3-round interactive proofs, with the aim of reducing the knowledge error while limiting the increase in the overall response size, we consider the t -fold parallel repetition of the protocol $(\mathcal{P}, \mathcal{V})$, where exactly w repetitions use the unfavourable challenge $\tilde{c} \in \text{Ch}^{[\mu]}$ in the last round. $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ is the resulting protocol. To show that this interactive proof is knowledge sound, we again start by slightly generalising the notation and results in [AF22].

A dishonest prover against $(\mathcal{P}, \mathcal{V})$ can be described as an arbitrary (probabilistic) algorithm $\mathcal{A}: \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]} \rightarrow \{0, 1\}^*$. Let $\mathbf{V}: \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]} \times \{0, 1\}^* \rightarrow \{0, 1\}$ be a verification function and $\mathcal{D} = (\mathcal{D}^{[1]}, \dots, \mathcal{D}^{[\mu]})$ a collection of probability distributions, where $\mathcal{D}^{[i]}$ is over $D^{[i]} \subseteq \text{Ch}^{[i]}$ with $|D^{[i]}| \geq k_i$. We define the $\mathcal{D}$ -success probability of $\mathcal{A}$ as:

$$\varepsilon^V(\mathcal{A}, \mathcal{D}) = \Pr[V(C, \mathcal{A}(C)) = 1],$$

where $C = (C^{[1]}, \dots, C^{[\mu]})$ is a random variable, with $C^{[i]}$ being distributed as $\mathcal{D}^{[i]}$. If C is uniformly distributed over $\text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$, we write $\varepsilon^V(\mathcal{A})$. Similarly, we adapt the punctured success probability of $\mathcal{A}$ as:

$$\delta_{\mathbf{k}}^V(\mathcal{A}, \mathcal{D}) = \min_{S^{[1]}, S^{[2]}(\cdot), \dots, S^{[\mu]}(\cdot)} \Pr[V(C, \mathcal{A}(C)) = 1 \mid \begin{matrix} C^{[1]} \notin S^{[1]} \wedge C^{[2]} \notin S^{[2]}(C^{[1]}) \wedge \dots \\ \dots \wedge C^{[\mu]} \notin S^{[\mu]}(C^{[1]}, \dots, C^{[\mu-1]}) \end{matrix}],$$

where the minimum is over all sets $S^{[1]} \in \text{Ch}^{[1]}|_{k_1-1}$, and over all functions $S^{[i]}: \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[i-1]} \rightarrow \text{Ch}^{[i]}|_{k_i-1}$, with $i = 2, \dots, \mu$. Here, for any $i \in \{1, \dots, \mu\}$, $\text{Ch}^{[i]}|_{k_i-1}$ denotes the set of subsets of $\text{Ch}^{[i]}$ with cardinality $k_i - 1$.

Next, we define an extraction algorithm $\mathcal{E}^{\mathcal{A}}(\mathcal{D})$ with oracle access to $\mathcal{A}$ that samples the challenges according to the distribution $\mathcal{D}$. Building on [AF22, Lemma 4], it is possible to show that $\mathcal{E}^{\mathcal{A}}(\mathcal{D})$ runs in expected polynomial time and succeeds with probability at least $\delta_{\mathbf{k}}^V(\mathcal{A}, \mathcal{D})/K$.

Lemma 7. *Let $k_1, \dots, k_\mu \in \mathbb{N}^*$, $K = \prod_{i=1}^\mu k_i$, $\text{Ch}^{[1]}, \dots, \text{Ch}^{[\mu]}$ be finite sets with $\text{Ch}^{[j]}$ having cardinality $N_j \geq k_j$, $V: \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]} \times \{0, 1\}^* \rightarrow \{0, 1\}$ an arbitrary function and $\mathcal{D} = (\mathcal{D}^{[1]}, \dots, \mathcal{D}^{[\mu]})$ a collection of probability distributions $\mathcal{D}^{[j]}$ with support equal to $\text{Ch}^{[j]}$. Then, there exists an algorithm $\mathcal{E}^{\mathcal{A}}(\mathcal{D})$ that, given oracle access to a (probabilistic) algorithm $\mathcal{A}: \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]} \rightarrow \{0, 1\}^*$, with an expected number of at most $2^\mu K$ queries to $\mathcal{A}$ and with probability at least $\delta_{\mathbf{k}}^V(\mathcal{A}, \mathcal{D})/K$, it outputs K pairs $(\mathbf{c}_1, y_1), \dots, (\mathbf{c}_K, y_K) \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]} \times \{0, 1\}^*$ with $V(\mathbf{c}_i, y_i) = 1$ for all $i \in \{1, \dots, K\}$ and such that the vectors $\mathbf{c}_i$ form a $(k_1, \dots, k_\mu)$ -tree of transcripts.*

Proof. The proof resembles that of [AF22, Lemma 4], with the only difference that the single-instance extractor used internally is an instantiation of the one described in Figure 2. $\square$

5.1 Knowledge-Soundness of Fixed-Weight Repetitions

Let $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ be the (t, w) -fixed-weight repetition – with respect to an unfavourable challenge $\tilde{c} \in \text{Ch}^{[\mu]}$ – of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proof $(\mathcal{P}, \mathcal{V})$ with challenge space $\text{Ch} = \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$.

For ease of notation, in the following we assume that the challenge space for the i -th round of $(\mathcal{P}, \mathcal{V})$ is $\text{Ch}^{[i]} = \{0, \dots, N_i - 1\}$ while the unfavourable challenge for the last round is $\tilde{c} = 0$.

Definition 22. *For each $j \in \{1, \dots, \mu - 1\}$, let $\mathcal{U}^{[j]}$ be the uniform distribution over $\text{Ch}^{[j]}$. For every $i \in \{1, \dots, t\}$, let $\mathcal{D}_i^{[\mu]}$ be the probability distribution having the following density function:*

$$\Pr[X_i = a] = \frac{|\{c \in (\text{Ch}^{[\mu]})_0^{t,w} \mid (c)_i = a\}|}{|(\text{Ch}^{[\mu]})_0^{t,w}|}, \quad \text{for all } a \in \text{Ch}^{[\mu]}.$$

Finally, let $\mathcal{D}_i = (\mathcal{U}^{[1]}, \dots, \mathcal{U}^{[\mu-1]}, \mathcal{D}_i^{[\mu]})$.

An adversary against $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ is described as a (possibly-probabilistic) algorithm which, on input a row $\mathbf{c} = ((\mathbf{c})_1, \dots, (\mathbf{c})_t)$ of columns $(\mathbf{c})_i = ((c)_i^{[1]}, \dots, (c)_i^{[\mu]}) \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$ of challenges such that $\text{wt}_0((c)_1^{[\mu]}, \dots, (c)_t^{[\mu]}) = w$, outputs a string $y \in \{0, 1\}^*$. The success probability of $\mathcal{A}$ is defined as:

$$\varepsilon^V(\mathcal{A}) = \Pr[\mathbf{V}(C, \mathcal{A}(C)) = 1],$$

for some verification algorithm $\mathbf{V}: \text{Ch}_0^{t,w} \times \{0, 1\}^* \rightarrow \{0, 1\}$, with C being a random variable uniformly distributed over $\text{Ch}_0^{t,w}$.

Such an algorithm $\mathcal{A}$ induces t algorithms $\mathcal{A}_1, \dots, \mathcal{A}_t$, analogous to those considered in the context of a single instance of $(\mathcal{P}, \mathcal{V})$. Each $\mathcal{A}_i$ takes as input a column $(\mathbf{c})_i \in \text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$. Then $\mathcal{A}_i$ runs $y \leftarrow \mathcal{A}(\mathbf{c} = ((\mathbf{c})_i, \bar{\mathbf{c}}))$, where $\bar{\mathbf{c}}$ is sampled uniformly at random from $\text{Ch}_0^{t-1, w-1}$ if $(c)_i^{[\mu]} = 0$ or from $\text{Ch}_0^{t-1, w}$ otherwise, and $\mathcal{A}$ is understood to appropriately reorder its input so that $(\mathbf{c})_i$ is the i -th component of $\mathbf{c}$. Finally $\mathcal{A}_i$ returns y along with $\bar{\mathbf{c}}$.

Notice that, when the input challenge for $\mathcal{A}_i$ is sampled according to the probability distribution $\mathcal{D}_i = (\mathcal{U}^{[1]}, \dots, \mathcal{U}^{[\mu-1]}, \mathcal{D}_i^{[\mu]})$ over $\text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$, then the inputs passed to $\mathcal{A}$ are uniformly distributed over $\text{Ch}_0^{t,w}$. Hence, for each $\mathcal{A}_i$, we can consider the extractor $\mathcal{E}^{\mathcal{A}_i}(\mathcal{D}_i)$ of Lemma 7, which succeeds with probability at least $\delta_{\mathbf{k}}^V(\mathcal{A}_i, \mathcal{D}_i)/K$, where:

$$\delta_{\mathbf{k}}^V(\mathcal{A}_i, \mathcal{D}_i) = \min_{S_i^{[1]}, S_i^{[2]}(\cdot), \dots, S_i^{[\mu]}(\cdot)} \Pr \left[\mathbf{V}(D_i, \mathcal{A}_i(D_i)) = 1 \mid \begin{array}{l} D_i^{[1]} \notin S_i^{[1]} \wedge D_i^{[2]} \notin S_i^{[2]} \wedge \dots \wedge \\ \dots \wedge D_i^{[\mu]} \notin S_i^{[\mu]}(D_i^{[1]}, \dots, D_i^{[\mu-1]}) \end{array} \right],$$

and D_i is distributed as $\mathcal{D}_i$.

In the following lemma we show that, when executed in parallel, at least one of the extractors $\mathcal{E}^{\mathcal{A}_i}(\mathcal{D}_i)$ succeeds with high probability in producing $\prod_{i=1}^{\mu} k_i$ challenge-response pairs that verify $\mathbf{V}$ and such that the challenges form a $(k_1, \dots, k_{\mu})$ -tree of transcripts.

Lemma 8. *Let $k_1, \dots, k_{\mu}, t, w \in \mathbb{N}^*$ such that $1 \leq w \leq t$ and let $\text{Ch}^{[1]}, \dots, \text{Ch}^{[\mu]}$ be finite sets with $\text{Ch}^{[j]}$ having cardinality $N_j \geq k_j$. Let $\mathbf{V}: \text{Ch}_0^{t,w} \times \{0, 1\}^* \rightarrow \{0, 1\}$ and let $\mathcal{A}$ be a (probabilistic) algorithm that takes as input an element $\mathbf{c}$ of $\text{Ch}_0^{t,w}$ and outputs a string $y \in \{0, 1\}^*$. Then:*

$$\sum_{i=1}^t \delta_{\mathbf{k}}^V(\mathcal{A}_i, \mathcal{D}_i) \geq \frac{\varepsilon^V(\mathcal{A}) - \kappa_{t,w}}{1 - \bar{\kappa}},$$

where $\kappa_{t,w}$ is the maximum, taken over $\alpha \in \{0, \dots, t\}$, of the expression:

$$\frac{\sum_{\ell=\max(0, w-t+\alpha)}^{\min(w, \alpha)} \binom{\alpha}{\ell} \binom{t-\alpha}{w-\ell} Z_0^{\ell} (Z_1 - Z_0)^{\alpha-\ell} (Z_2)^{w-\ell} (Z_1 - Z_2)^{t-\alpha-w+\ell}}{|\text{Ch}_0^{t,w}|},$$

with $|\text{Ch}_0^{t,w}| = \binom{t}{w} (N_\mu - 1)^{t-w} (\prod_{i=1}^{\mu-1} N_i)^t$ and

$$\begin{aligned} Z_0 &:= \prod_{\ell=1}^{\mu-1} N_\ell, \\ Z_1 &:= \sum_{\ell=1}^{\mu} \left(\prod_{j=\ell+1}^{\mu} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right), \\ Z_2 &:= \sum_{\ell=1}^{\mu-1} \left(\prod_{j=\ell+1}^{\mu-1} N_j \right) (k_\ell - 1) \left(\prod_{j=1}^{\ell-1} (N_j - k_j + 1) \right). \end{aligned}$$

Moreover, it holds that:

$$\bar{\kappa} \geq 1 - \left(\prod_{j=1}^{\mu-1} \frac{N_j - k_j + 1}{N_j} \right) \left(\frac{N_\mu - k_\mu + 1}{N_\mu - 1} \frac{t - w}{w} + \frac{w}{t} \right).$$

Proof. Let $(C)_i$ be the i -th component of the random variable C uniformly distributed over $\text{Ch}_0^{t,w}$ and let Λ be the event $\mathbb{V}(C, \mathcal{A}(C)) = 1$. Therefore $\Pr[\Lambda] = \varepsilon^V(\mathcal{A})$. For $i \in \{1, \dots, t\}$, let $S_i^{[1]}$ and $S_i^{[2]}(\cdot), \dots, S_i^{[\mu]}(\cdot)$ be such that they minimize $\delta_{\mathbf{k}}^V(\mathcal{A}_i, \mathcal{D}_i)$. Moreover, we denote by Γ_i the event:

$$(C)_i^{[1]} \notin S_i^{[1]} \wedge (C)_i^{[2]} \notin S_i^{[2]}((C)_i^{[1]}) \wedge \dots \wedge (C)_i^{[\mu-1]} \notin S_i^{[\mu-1]}((C)_i^{[1]}, \dots, (C)_i^{[\mu-2]}),$$

and by Ω_i the event $(C)_i^{[\mu]} \notin S_i^{[\mu]}((C)_i^{[1]}, \dots, (C)_i^{[\mu-1]})$. Furthermore, we consider the probability distribution D_i which is distributed as $\mathcal{D}_i = (\mathcal{D}_i^{[1]}, \dots, \mathcal{D}_i^{[\mu-1]}, \mathcal{D}_i^{[\mu]})$ over $\text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$. Therefore, similarly to the proof of Lemma 3, D_i and $(C)_i$ are identically distributed for any $i \in \{1, \dots, t\}$ and, by construction of the $\mathcal{A}_i$'s, it holds that:

$$\begin{aligned} \sum_{i=1}^t \delta_{\mathbf{k}}^V(\mathcal{A}_i, \mathcal{D}_i) &= \sum_{i=1}^t \Pr \left[\mathbb{V}(D_i, \mathcal{A}_i(D_i)) = 1 \mid \begin{array}{l} D_i^{[1]} \notin S_i^{[1]} \wedge D_i^{[2]} \notin S_i^{[2]}(D_i^{[1]}) \wedge \dots \\ \dots \wedge D_i^{[\mu-1]} \notin S_i^{[\mu-1]}(D_i^{[1]}, \dots, D_i^{[\mu-2]}) \end{array} \right] \\ &= \sum_{i=1}^t \Pr[\Lambda \mid \Gamma_i \cap \Omega_i]. \end{aligned}$$

From elementary probability, it follows that:

$$\begin{aligned} \sum_{i=1}^t \Pr[\Lambda \mid \Gamma_i \cap \Omega_i] &= \sum_{i=1}^t \frac{\Pr[\Lambda \wedge (\Gamma_i \cap \Omega_i)]}{\Pr[\Gamma_i \cap \Omega_i]} = \sum_{i=1}^t \frac{\Pr[\Lambda \wedge (\Gamma_i \cap \Omega_i)]}{\Pr[\Gamma_1 \cap \Omega_1]} \\ &\geq \frac{\Pr[\Lambda \wedge \bigcup_i (\Gamma_i \cap \Omega_i)]}{1 - \bar{\kappa}} \geq \frac{\Pr[\Lambda] - \Pr[\bigcap_i \overline{(\Gamma_i \cap \Omega_i)}]}{1 - \bar{\kappa}}, \end{aligned}$$

where in the first inequality we can take $1/(1 - \bar{\kappa})$ out of the sum by defining $\bar{\kappa} = 1 - \Pr[\Gamma_1 \cap \Omega_1]$ and observing that $\Pr[\Gamma_1 \cap \Omega_1] = \dots = \Pr[\Gamma_t \cap \Omega_t]$. To obtain the expression of $\bar{\kappa}$, first write $\Pr[\Gamma_1 \cap \Omega_1] = \Pr[\Gamma_1] \cdot \Pr[\Omega_1 \mid \Gamma_1]$ and notice that:

$$\Pr[\Gamma_1] = \prod_{j=1}^{\mu-1} \frac{N_j - k_j + 1}{N_j}.$$

Moreover, observe that $\Pr[\Omega_1 \mid I_1] = \Pr[(C)_1^{[\mu]} \notin S_1^{[\mu]}]$ for some set $S_1^{[\mu]} \subset \text{Ch}^{[\mu]}$ with $|S_1^{[\mu]}| = k_\mu - 1$. Now, let $\bar{S}_1^{[\mu]} = \text{Ch}^{[\mu]} \setminus S_1^{[\mu]}$, then:

$$\Pr[(C)_1^{[\mu]} \notin S_1^{[\mu]}] = \Pr[(C)_1^{[\mu]} \in \bar{S}_1^{[\mu]}] = \frac{\bar{\eta}}{|(\text{Ch}^{[\mu]})_0^{t,w}|},$$

where the value of $\bar{\eta}$ is given by the size of a maximal $\bar{S} \subseteq (\text{Ch}^{[\mu]})_0^{t,w}$ such that $|(\bar{S})_1| \leq N_\mu - k_\mu + 1$. Following the same techniques of Proposition 1, we can explicitly compute $\bar{\eta}$ depending on whether $0 \in \bar{S}_1^{[\mu]}$. It holds that:

$$\begin{aligned} \bar{\eta} &= \max \left\{ \begin{array}{ll} \binom{t-1}{w} (N_\mu - k_\mu + 1) (N_\mu - 1)^{t-w-1} & \text{If } 0 \notin \bar{S}_1^{[\mu]} \\ \binom{t-1}{w} (N_\mu - k_\mu) (N_\mu - 1)^{t-w-1} + \binom{t-1}{w-1} (N_\mu - 1)^{t-w} & \text{If } 0 \in \bar{S}_1^{[\mu]} \end{array} \right\} \\ &\leq \binom{t-1}{w} (N_\mu - k_\mu + 1) (N_\mu - 1)^{t-w-1} + \binom{t-1}{w-1} (N_\mu - 1)^{t-w} \\ &= \binom{t}{w} (N_\mu - 1)^{t-w} \left(\frac{N_\mu - k_\mu + 1}{N_\mu - 1} \frac{t-w}{w} + \frac{w}{t} \right). \end{aligned}$$

Since $|(\text{Ch}^{[\mu]})_0^{t,w}| = \binom{t}{w} (N_\mu - 1)^{t-w}$, we obtain:

$$\Pr[I_1 \cap \Omega_1] \leq \left(\prod_{j=1}^{\mu-1} \frac{N_j - k_j + 1}{N_j} \right) \left(\frac{N_\mu - k_\mu + 1}{N_\mu - 1} \frac{t-w}{w} + \frac{w}{t} \right).$$

Now, let $\kappa_{t,w} = \Pr[\bigcap_{i=1}^t (\bar{I}_i \cap \Omega_i)]$. For any $i \in \{1, \dots, t\}$, the set $S_i^{[1]}$ and the maps $S_i^{[2]}(\cdot), \dots, S_i^{[\mu]}(\cdot)$ identify a subset $(\bar{A})_i$ of $\text{Ch}^{[1]} \times \dots \times \text{Ch}^{[\mu]}$ defined in the following way. The set $S_i^{[1]}$ dictates that $(\bar{A})_i$ contains:

$$\left\{ (c^{[1]}, c^{[2]}, \dots, c^{[\mu]}) : c^{[1]} \in S_i^{[1]} \wedge (c^{[2]}, \dots, c^{[\mu]}) \in \prod_{j=2}^{\mu} \text{Ch}^{[j]} \right\}.$$

Furthermore, the set $S_i^{[1]}$ and the map $S_i^{[2]}(\cdot)$ dictate that $(\bar{A})_i$ also contains:

$$\left\{ (c^{[1]}, c^{[2]}, \dots, c^{[\mu]}) : c^{[1]} \notin S_i^{[1]} \wedge c^{[2]} \in S_i^{[2]}(c^{[1]}) \wedge (c^{[3]}, \dots, c^{[\mu]}) \in \prod_{j=3}^{\mu} \text{Ch}^{[j]} \right\}.$$

By iterating this argument, we deduce that the sets $(\bar{A})_i$ have a form identical to the sets in the proof of Lemma 4 which have the same names, the only difference being that within that proof the sets $(\bar{A})_i$ were determined by a starting acceptable set A , while here they are determined by $S_i^{[1]}$ and $S_i^{[2]}(\cdot), \dots, S_i^{[\mu]}(\cdot)$. As a consequence, $\kappa_{t,w}$ corresponds to the probability of belonging to the set $\bar{A}$, which is defined as the set containing every element $((y)_1, \dots, (y)_t) \in \prod_{i=1}^t (\bar{A})_i$ which respects the first condition of Definition 17, i.e. $((y)_1^{[\mu]}, \dots, (y)_t^{[\mu]}) \in (\text{Ch}^{[\mu]})_0^{t,w}$. In other words, $\bar{A}$ is a saturated acceptable set, as defined in Definition 20, and then we can apply Proposition 2 to conclude that the probability $\kappa_{t,w}$ is exactly the one in the claim. $\square$

As for the fixed-weight repetition of a Sigma protocol, following Lemma 8 we can bound the probability that at least one extractor $\mathcal{E}^{\mathcal{A}_i}(\mathcal{D}_i)$ is successful:

$$\max_{1 \leq i \leq t} \delta_{\mathbf{k}}^{\mathcal{V}}(\mathcal{A}_i, \mathcal{D}_i) \geq \frac{1}{t} \sum_{i=1}^t \delta_{\mathbf{k}}^{\mathcal{V}}(\mathcal{A}_i, \mathcal{D}_i) \geq \frac{\varepsilon^{\mathcal{V}}(\mathcal{A}) - \kappa_{t,w}}{t \cdot (1 - \bar{\kappa})}.$$

It follows that the (t, w) -fixed-weight repetition $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proof $(\mathcal{P}, \mathcal{V})$ is knowledge sound with knowledge error $\kappa_{t,w}$, which again corresponds to the maximum cheating probability of a dishonest prover.

Theorem 2 (Fixed-Weight Repetition of a $(k_1, \dots, k_\mu)$ -Special-Sound Multi-Round Interactive Proof). *Let $(\mathcal{P}, \mathcal{V})$ be a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proof and $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ be the (t, w) -fixed-weight repetition of $(\mathcal{P}, \mathcal{V})$, where $w, k, t \in \mathbb{N}^*$ and $1 \leq w \leq t$. Then $(\mathcal{P}^{t,w}, \mathcal{V}^{t,w})$ is knowledge sound with knowledge error $\kappa_{t,w}$, where $\kappa_{t,w}$ is the maximum, taken over $\alpha \in \{0, \dots, t\}$, of the expression*

$$\frac{\sum_{\ell=\max(0, w-t+\alpha)}^{\min(w, \alpha)} \binom{\alpha}{\ell} \binom{t-\alpha}{w-\ell} Z_0^\ell (Z_1 - Z_0)^{\alpha-\ell} (Z_2)^{w-\ell} (Z_1 - Z_2)^{t-\alpha-w+\ell}}{\binom{t}{w} (N_\mu - 1)^{t-w} (\prod_{i=1}^{\mu-1} N_i)^t},$$

where Z_0, Z_1, Z_2 are defined as in Lemma 8.

Remark 6. Again, from the proof of Lemma 8, it follows that the knowledge error $\kappa_{t,w}$ coincides with the maximum cheating probability $\Pr[\bigcap_i \overline{\Gamma_i} \cap \Omega_i]$ which is potentially achievable by a dishonest prover against the (t, w) -fixed-weight repetition of a $(k_1, \dots, k_\mu)$ -special-sound $(2\mu + 1)$ -round interactive proof.

6 Applications and Conclusions

In this paper, we have established a positive result about the security of fixed-weight parallel repetitions of special-sound (multi-round) interactive proofs. We have obtained a strong result on the knowledge soundness of the fixed-weight optimization, proving that the knowledge error of the protocol matches the maximum cheating probability of a dishonest prover, for which we have provided an explicit expression. To the best of our knowledge, this is the first time the security of this standard optimization has been analysed, beyond 2-special-sound Sigma protocols.

Our work gives direct, tight results on the security of the interactive proofs underlying many recent signatures. For instance, they provide an explicit knowledge error for the fixed-weight repetition of $q2$ -identification schemes [Che+16], such as the 5-round interactive proof underlying CROSS [Bal+23]. Similarly, they can be applied to k -special-sound Sigma protocols, with $k > 2$, such as the recent SIDH-based signature of [GPV24].

Future Works. When dealing with multi-round interactive proofs, our results cover the fixed-weight optimization in the case where a fixed challenge in the last round is repeated a prescribed number of times. This is a seemingly arbitrary choice, as we might consider fixed-weight challenges in intermediate rounds or in a generic subset of them, but it is closely tailored to concrete applications of interactive proofs for building digital signatures. Indeed, the fixed-weight optimization is motivated by the presence of challenges with larger response sizes, while intermediate rounds have typically constant-size responses. However, an extension of our results to a “generalized” fixed-weight optimization might be of interest for future protocols with different approaches from the current ones.

On the negative side, our result does not directly translate to the non-interactive case and thus we cannot use the bound of Sections 3 and 5 to determine parameters of signature schemes. Indeed, when considering signature schemes, it is necessary to also take into consideration the security loss caused by the Fiat-Shamir Transform. As shown in [AFK22], the Fiat-Shamir transform of a t -fold parallel repetition of a $(k_1, \dots, k_\mu)$ -special-sound interactive proof incurs in a security loss that is exponential in the number of rounds. While the attack of [AFK22] does not specifically target fixed-weight repetitions, the same heuristic could be applied to our scenario. Adapting the attack to find precise bounds for the security loss is an interesting open problem for future research, since they are crucial to determine provable-secure parameters of multi-round-based signatures, such as CROSS [Bal+23].

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